From Random Walks to Brownian Motion

Martingales, Lévy Continuity, and Donsker Scaling, Wiener Measure, Girsanov, and PDE Methods

Formal notes in the NYU Courant visual style

Discrete-continuous spine

For the symmetric random walk $S_n = \sum_{j=1}^{n} \xi_j$,

$$\mathbb{E}[S_n] = 0, \qquad \mathrm{Var}(S_n) = n, \qquad [S]_n = \langle S \rangle_n = n,$$

and $e^{\theta S_n}/(\cosh\theta)^n$ is a martingale. Under diffusion scaling these objects converge to their Brownian counterparts. In the drifted model $X_t = x_0 + \mu t + \sigma B_t$, the cumulant rate of the exponential $e^{\theta(X_t - x_0)}$ is

$$\psi(\theta) = \mu\theta + \frac{1}{2}\sigma^2\theta^2.$$

Hence the compensated process

$$\exp(\theta(X_t - x_0) - \psi(\theta)t)$$

is a mean-one martingale for every $\theta \in \mathbb{R}$. The uncorrected exponential is a martingale at the nontrivial calibration $\theta_* = -2\mu/\sigma^2$.

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September 7, 2026

Abstract

These notes develop the discrete-to-continuous architecture used in a Courant treatment of stochastic processes. Rademacher increments generate the symmetric random walk, its elementary martingales, optional-stopping formulas, and diffusion scaling. Characteristic functions are separated from moment generating functions, and Lévy's continuity theorem is distinguished from pointwise, uniform, in-measure, and path-space convergence. Cantor devil's-staircase constructions make atoms, tightness, and escaped mass visible. The analytic half derives the Brownian heat kernel by Fourier inversion and an entire-function contour shift. The measure-theoretic half then constructs Wiener measure from consistent Gaussian finite-dimensional laws, obtains continuous paths through Kolmogorov–Chentsov, builds the stochastic integral by completion in L^2, and proves Itô isometry and Itô's formula. A complete Donsker proof establishes tightness in $C([0,T])$ by dyadic modulus estimates and Arzelà–Ascoli before applying Prokhorov. Finally, Girsanov's theorem identifies the Radon–Nikodym density for drift removal, while the generator, Dynkin formula, Feynman–Kac formula, and distributional forward equation connect the probabilistic construction to classical, mild, and weak PDE solutions. Five Python figures retain a common NYU-violet notation throughout.

Contents

1 Model, notation, and transform conventions **1**
1.1 Characteristic functions versus moment generating functions . . . 1

2 Symmetric random walk: the discrete skeleton **2**
2.1 The Rademacher increment . . . 2
2.2 Definition, expectation, variance, and covariance . . . 2
2.3 Three elementary martingales . . . 3
2.4 Optional stopping in discrete time . . . 3
2.5 Optional and predictable quadratic variation . . . 4
2.6 Diffusion scaling and weak drift bias . . . 4
2.7 Discrete exponential calibration and its continuous limit . . . 5
2.8 Gambler's ruin and the Brownian exit formula . . . 5

3 Gaussian MGF and the substitution $k = \theta\sigma$ **6**
3.1 Direct completion-of-the-square calculation . . . 6
3.2 Why introduce $k = \theta\sigma$? . . . 7

4 Exponential-martingale calibration **7**

5 Characteristic function and Brownian transition density **8**
5.1 Fourier inversion . . . 8
5.2 The entire-function contour shift . . . 8
5.3 Gaussian normalization and the heat kernel . . . 9

6 Two-sided absorption via optional stopping **9**

7 Python visualizations of the discrete–continuous limit **11**

8 Modes of convergence **14**
8.1 Pointwise, uniform, almost-everywhere, and in-measure convergence . . . 14
8.2 Sharp counterexamples and the Riesz subsequence theorem . . . 14
8.3 Equicontinuity and the Arzelà–Ascoli bridge . . . 15

9 Lévy continuity and the passage from laws to paths **15**
9.1 The exact continuity theorem . . . 15
9.2 A compressed Cantor staircase: a CDF jump is not a CF discontinuity . . . 16

9.3 A partially escaping staircase: discontinuity at frequency zero . . . 17
9.4 Tightness, Prokhorov compactness, and local uniformity . . . 20
9.5 Weak convergence does not remember the coupling . . . 21
9.6 Fixed-time convergence of the weakly biased walk . . . 22
9.7 Donsker's theorem supplies tightness . . . 22
9.8 A second theorem bearing Lévy's name . . . 23

10 Measure-theoretic construction of Wiener measure **24**
10.1 Consistent Gaussian laws and the extension theorem . . . 24
10.2 Kolmogorov–Chentsov and the continuous modification . . . 25

11 The L^2 Itô integral and Itô isometry **26**
11.1 Elementary predictable integrands . . . 26
11.2 Completion in L^2 . . . 26
11.3 Itô's formula as the stochastic chain rule . . . 27

12 A complete path-space proof of Donsker's theorem **28**
12.1 Finite-dimensional convergence . . . 28
12.2 The uniform fourth-moment estimate . . . 28
12.3 Dyadic modulus control and Arzelà–Ascoli . . . 28
12.4 Prokhorov plus uniqueness of the subsequential limit . . . 29

13 Girsanov change of measure and drift calibration **30**
13.1 The constant-drift theorem with a complete calculation . . . 30
13.2 General predictable drift . . . 30

14 Generator, Feynman–Kac, and weak PDE solutions **32**
14.1 Infinitesimal generator and Dynkin's formula . . . 32
14.2 Backward equation and Feynman–Kac . . . 32
14.3 Forward equation as a weak identity . . . 33

15 Dependency map of the complete argument **34**

1 Model, notation, and transform conventions

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, and let $(B_t)_{t\geq 0}$ be standard Brownian motion. Fix constants

$$x_0 \in \mathbb{R}, \qquad \mu \in \mathbb{R}, \qquad \sigma > 0,$$

and define the drifted, volatility-scaled process

$$X_t = x_0 + \mu t + \sigma B_t. \tag{1.1}$$

At a fixed time $t > 0$,

$$B_t \sim N(0, t), \qquad X_t \sim N(x_0 + \mu t, \sigma^2 t).$$

We use the Fourier convention

$$\widehat{f}(\omega) = \int_{\mathbb{R}} e^{i\omega x} f(x)\, \mathrm{d}x, \qquad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\omega x} \widehat{f}(\omega)\, \mathrm{d}\omega. \tag{1.2}$$

For a real random variable Y, its moment generating function and characteristic function are respectively

$$M_Y(k) = \mathbb{E}[e^{kY}], \qquad \phi_Y(\omega) = \mathbb{E}[e^{i\omega Y}].$$

1.1 Characteristic functions versus moment generating functions

Both transforms encode a probability law, but they live in different analytic regimes. The characteristic function is the Fourier–Stieltjes transform of the law:

$$\phi_Y(u) = \int_{\mathbb{R}} e^{iux}\, \mathbb{P}_Y(\,\mathrm{d}x).$$

Since $|e^{iuY}| = 1$, it exists for every real-valued random variable and every $u \in \mathbb{R}$. The MGF is the bilateral Laplace transform

$$M_Y(k) = \int_{\mathbb{R}} e^{kx}\, \mathbb{P}_Y(\,\mathrm{d}x),$$

and can be infinite. Its effective domain is

$$\mathcal{D}_Y := \{k \in \mathbb{R} : M_Y(k) < \infty\}. \tag{1.3}$$

Although $0 \in \mathcal{D}_Y$ always, the phrase *the MGF exists* usually means that M_Y is finite on an open interval containing zero.

Table 1: Characteristic function and moment generating function.

Feature	Characteristic function $\phi_Y(u)$	Moment generating function $M_Y(k)$
Kernel	e^{iux}, oscillatory and bounded in modulus by 1	e^{kx}, a real exponential weight
Existence	Finite for every $u \in \mathbb{R}$ and every probability law	May be infinite for every nonzero k, or finite only on one side of zero
Transform	Fourier–Stieltjes transform of the probability measure	Bilateral Laplace transform of the probability measure
Moments	If $\mathbb{E}\|Y\|^m < \infty$, then $\phi_Y^{(m)}(0) = i^m \mathbb{E}[Y^m]$	If finite in a neighborhood of 0, then $M_Y^{(m)}(0) = \mathbb{E}[Y^m]$
Uniqueness	Always determines the probability law uniquely	Determines the law when finite on an open interval about 0
Geometry	Maps Y to rotations on the complex unit circle	Exponentially amplifies one tail and suppresses the other

Where analytic continuation is legitimate, the two transforms are related by

$$M_Y(k) = \phi_Y(-ik). \tag{1.4}$$

This substitution is not formal license to evaluate a characteristic function at complex arguments: the expectation on the right must remain finite.

Remark 1.1 (Existence and inversion qualifications)**.** A Cauchy random variable has no finite MGF at any $k \neq 0$. Positive Pareto and lognormal variables have $M_Y(k) = \infty$ for every $k > 0$, so neither has an MGF on a two-sided neighborhood of zero. Their characteristic functions nevertheless exist everywhere. Also, the characteristic function always determines the law, but the elementary density inversion integral requires additional hypotheses such as $\phi_Y \in L^1(\mathbb{R})$. Without a density one uses the general Fourier–Stieltjes inversion theorem for distribution functions.

Remark 1.2 (Avoiding a circular derivation)**.** If Brownian motion is defined as a Gaussian process with independent stationary increments, then $B_t \sim N(0,t)$ is part of the definition and Fourier inversion merely reconstructs the density. If one starts from the generator $\frac{1}{2}\partial_{xx}$, the heat equation, or a continuous local martingale with quadratic variation t, then the characteristic function can first be derived independently and inverted afterward.

2 Symmetric random walk: the discrete skeleton

2.1 The Rademacher increment

Definition 2.1 (Rademacher random variable)**.** A random variable R is Rademacher when

$$\mathbb{P}(R=1) = \mathbb{P}(R=-1) = \frac{1}{2}, \qquad \mathcal{L}(R) = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1. \tag{2.1}$$

Its symmetry makes every odd moment vanish, while every even moment is one:

$$\mathbb{E}[R^{2m+1}] = 0, \qquad \mathbb{E}[R^{2m}] = 1 \qquad (m \geq 0).$$

In particular,

$$\mathbb{E}[R] = 0, \qquad \operatorname{Var}(R) = 1.$$

The two transforms display the Fourier/Laplace distinction in its simplest form:

$$\phi_R(u) = \frac{e^{iu} + e^{-iu}}{2} = \cos u, \tag{2.2}$$

$$M_R(k) = \frac{e^{k} + e^{-k}}{2} = \cosh k. \tag{2.3}$$

Thus the CF oscillates in $[-1,1]$, whereas the MGF grows exponentially. If $A \sim \text{Bernoulli}(1/2)$, then

$$R = 2A - 1. \tag{2.4}$$

For independent copies $R_1, \ldots, R_n$, the normalized sum already contains the characteristic-function proof of the central limit theorem:

$$\phi_{n^{-1/2}\sum_{j=1}^n R_j}(u) = \left(\cos\frac{u}{\sqrt{n}}\right)^n \tag{2.5}$$

$$= \exp\left[n \log\cos\left(\frac{u}{\sqrt{n}}\right)\right] \longrightarrow e^{-u^2/2}, \tag{2.6}$$

locally uniformly in u, because $\log\cos z = -z^2/2 + O(z^4)$ near zero. Lévy's theorem then identifies the limit as $N(0,1)$.

2.2 Definition, expectation, variance, and covariance

Let $(\xi_j)_{j\geq 1}$ be independent Rademacher random variables, and define the symmetric random walk and its natural filtration by

$$S_0 = 0, \qquad S_n = \sum_{j=1}^n \xi_j, \qquad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n). \tag{2.7}$$

Since

$$\mathbb{E}[\xi_j] = 0, \qquad \mathbb{E}[\xi_j^2] = 1, \qquad \operatorname{Var}(\xi_j) = 1,$$

independence gives the exact first and second moments.

Proposition 2.2 (Moments of the SRW)**.** *For all integers $m, n \geq 0$,*

$$\mathbb{E}[S_n] = 0, \qquad \operatorname{Var}(S_n) = n, \qquad \operatorname{Cov}(S_m, S_n) = \min\{m,n\}. \tag{2.8}$$

In particular, $\mathbb{E}[S_n^2] = n$.

Proof. Linearity gives

$$\mathbb{E}[S_n] = \sum_{j=1}^{n} \mathbb{E}[\xi_j] = 0.$$

All cross-covariances vanish, so

$$\operatorname{Var}(S_n) = \sum_{j=1}^{n} \operatorname{Var}(\xi_j) = n.$$

If $m \le n$, write $S_n = S_m + (S_n - S_m)$. The increment $S_n - S_m$ is independent of S_m and centered; hence

$$\operatorname{Cov}(S_m, S_n) = \operatorname{Var}(S_m) = m.$$

Symmetry in m, n yields the final formula. □

2.3 Three elementary martingales

Proposition 2.3 (Linear, quadratic, and exponential martingales)**.** *For every $\theta \in \mathbb{R}$, the following adapted processes are martingales:*

$$S_n, \qquad S_n^2 - n, \qquad Z_n(\theta) := \frac{e^{\theta S_n}}{(\cosh \theta)^n}. \tag{2.9}$$

Proof. Because ξ_{n+1} is independent of $\mathcal{F}_n$ and centered,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[\xi_{n+1}] = S_n.$$

Likewise,

$$\begin{aligned}\mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] &= S_n^2 + 2S_n \mathbb{E}[\xi_{n+1}] + \mathbb{E}[\xi_{n+1}^2] - (n+1) \\ &= S_n^2 - n.\end{aligned}$$

Finally,

$$\mathbb{E}[e^{\theta \xi_{n+1}}] = \frac{e^{\theta} + e^{-\theta}}{2} = \cosh \theta,$$

and therefore

$$\mathbb{E}[Z_{n+1}(\theta) \mid \mathcal{F}_n] = \frac{e^{\theta S_n}}{(\cosh \theta)^{n+1}} \mathbb{E}[e^{\theta \xi_{n+1}}] = Z_n(\theta).$$

□

2.4 Optional stopping in discrete time

Theorem 2.4 (Optional stopping theorem, useful forms)**.** *Let $(M_n, \mathcal{F}_n)_{n \ge 0}$ be an integrable discrete-time martingale and τ a stopping time. For every deterministic N,*

$$\mathbb{E}[M_{\tau \wedge N}] = \mathbb{E}[M_0]. \tag{2.10}$$

If $\tau < \infty$ almost surely, moreover, $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ follows if, for example, any one of the following holds:

(i) τ is almost surely bounded;
(ii) the stopped family $\{M_{\tau \wedge N} : N \ge 1\}$ is uniformly integrable;
(iii) $\mathbb{E}[\tau] < \infty$ and the increments satisfy $|M_n - M_{n-1}| \le C$ almost surely for a deterministic C.

Core calculation. For bounded stopping,

$$M_{\tau \wedge N} = M_0 + \sum_{j=1}^{N} (M_j - M_{j-1}) \mathbf{1}_{\{\tau \ge j\}}.$$

Because $\{\tau \ge j\} \in \mathcal{F}_{j-1}$,

$$\mathbb{E}\big[(M_j - M_{j-1}) \mathbf{1}_{\{\tau \ge j\}}\big] = \mathbb{E}\big[\mathbf{1}_{\{\tau \ge j\}} \mathbb{E}[M_j - M_{j-1} \mid \mathcal{F}_{j-1}]\big] = 0.$$

Summation proves (2.10). Each additional hypothesis gives the integrability needed to pass from $\tau \wedge N$ to τ. □

Why OST needs hypotheses

The facts that M is a martingale and $\tau < \infty$ almost surely do not alone justify $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$. The theorem is a limit passage, and boundedness, uniform integrability, or an equivalent domination condition is what prevents expectation from being lost at large stopping times.

2.5 Optional and predictable quadratic variation

For a discrete local martingale M, distinguish the realized quadratic variation

$$[M]_n := \sum_{j=1}^{n} (\Delta M_j)^2 \tag{2.11}$$

from the predictable bracket

$$\langle M \rangle_n := \sum_{j=1}^{n} \mathbb{E}[(\Delta M_j)^2 \mid \mathcal{F}_{j-1}]. \tag{2.12}$$

For the SRW, $\Delta S_j = \xi_j$ and $\xi_j^2 = 1$ deterministically, so

$$\boxed{[S]_n = \langle S \rangle_n = n.} \tag{2.13}$$

For SRW the two quantities coincide numerically. Modern notation nevertheless reserves the realized square sum for $[M]_n$; for a general martingale, $[M]_n$ and $\langle M \rangle_n$ need not coincide.

2.6 Diffusion scaling and weak drift bias

Let $\delta = \Delta t > 0$ and choose the spatial step

$$h = \Delta x = \sigma\sqrt{\delta}. \tag{2.14}$$

For a weakly biased walk, define one-step increments by

$$\Delta X_j^\delta = \begin{cases} +h, & \text{with probability } p_\delta, \\ -h, & \text{with probability } q_\delta = 1 - p_\delta. \end{cases}$$

Matching the desired drift $\mathbb{E}[\Delta X_j^\delta] = \mu\delta$ gives

$$(2p_\delta - 1)\sigma\sqrt{\delta} = \mu\delta, \tag{2.15}$$

$$\boxed{p_\delta = \frac{1}{2}\left(1 + \frac{\mu}{\sigma}\sqrt{\delta}\right), \qquad q_\delta = \frac{1}{2}\left(1 - \frac{\mu}{\sigma}\sqrt{\delta}\right).} \tag{2.16}$$

The factor $1/2$ is simply the solution of the linear equation $2p_\delta - 1 = (\mu/\sigma)\sqrt{\delta}$: the bias is added around the symmetric baseline $1/2$. One must take δ small enough that $|\mu|\sqrt{\delta} \le \sigma$, ensuring $p_\delta, q_\delta \in [0, 1]$.

For $N = t/\delta \in \mathbb{N}$, let

$$X_t^\delta = \sum_{j=1}^{N} \Delta X_j^\delta.$$

The one-step and accumulated moments are

$$\mathbb{E}[\Delta X_j^\delta] = \mu\delta, \tag{2.17}$$

$$\mathrm{Var}(\Delta X_j^\delta) = \mathbb{E}[(\Delta X_j^\delta)^2] - \mu^2\delta^2 = \sigma^2\delta - \mu^2\delta^2, \tag{2.18}$$

$$\mathbb{E}[X_t^\delta] = \mu t, \tag{2.19}$$

$$\mathrm{Var}(X_t^\delta) = \sigma^2 t - \mu^2 t\delta \longrightarrow \sigma^2 t. \tag{2.20}$$

Moreover, the realized square sum is deterministic:

$$\sum_{j=1}^{N} (\Delta X_j^\delta)^2 = N\sigma^2\delta = \sigma^2 t. \tag{2.21}$$

With $N = \lfloor t/\delta \rfloor$, the discrepancy is at most $\sigma^2\delta$. Hence this quadratic-variation clock converges uniformly on compact time intervals to $\sigma^2 t$. This is an exact structural match; by itself it does not prove process-level convergence.

2.7 Discrete exponential calibration and its continuous limit

For scaled symmetric increments $\pm\sigma\sqrt{\delta}$, define

$$Z_t^\delta(\theta) = \frac{\exp(\theta X_t^\delta)}{\big(\cosh(\theta\sigma\sqrt{\delta})\big)^{t/\delta}}. \tag{2.22}$$

At lattice times this is a martingale. Since

$$\log\cosh y = \frac{y^2}{2} + O(y^4),$$

we obtain

$$\frac{t}{\delta}\log\cosh(\theta\sigma\sqrt{\delta}) = \frac{1}{2}\theta^2\sigma^2 t + O(t\delta). \tag{2.23}$$

Thus the denominator converges to $e^{\theta^2\sigma^2 t/2}$, and the limiting stochastic exponential is

$$\exp\left(\theta\sigma B_t - \frac{1}{2}\theta^2\sigma^2 t\right). \tag{2.24}$$

For the biased probabilities (2.16), the one-step MGF is

$$m_\delta(\theta) = p_\delta e^{\theta\sigma\sqrt{\delta}} + q_\delta e^{-\theta\sigma\sqrt{\delta}} \tag{2.25}$$

$$= 1 + \left(\mu\theta + \frac{1}{2}\sigma^2\theta^2\right)\delta + O(\delta^2), \tag{2.26}$$

locally uniformly in θ. Consequently,

$$m_\delta(\theta)^{t/\delta} \longrightarrow \exp\left[\left(\mu\theta + \frac{1}{2}\sigma^2\theta^2\right)t\right], \tag{2.27}$$

which is precisely the compensator appearing in the continuous exponential martingale.

2.8 Gambler's ruin and the Brownian exit formula

Let an integer-valued biased walk start at zero and stop on first hitting $-a$ or $+b$, where $a, b \in \mathbb{N}$. With $r = q/p$, the upper-hit probability is

$$\mathbb{P}_0(\tau_b < \tau_{-a}) = \begin{cases} \dfrac{1-r^a}{1-r^{a+b}}, & p \neq q, \\ \dfrac{a}{a+b}, & p = q = \frac{1}{2}. \end{cases} \tag{2.28}$$

The denominator $1-r^{a+b}$ is essential for boundaries $-a$ and $+b$; the expression $r^b - r^a$ corresponds to a different coordinate convention and cannot be inserted without translating the starting point and boundaries.

Martingale/OST derivation of (2.28). Let Y_n be the walk and $\tau = \inf\{n \geq 0 : Y_n \in \{-a, b\}\}$. When $p \neq q$, put $r = q/p$. Since

$$\mathbb{E}[r^{Y_{n+1}} \mid \mathcal{F}_n] = r^{Y_n}\big(pr + qr^{-1}\big) = r^{Y_n},$$

the process r^{Y_n} is a martingale. Before and at exit, $Y_{\tau\wedge n} \in [-a, b]$, so the stopped family is bounded. Theorem 2.4 gives, with $\pi = \mathbb{P}_0(Y_\tau = b)$,

$$1 = \mathbb{E}[r^{Y_\tau}] = \pi r^b + (1-\pi)r^{-a}.$$

Solving yields

$$\pi = \frac{1-r^a}{1-r^{a+b}}.$$

When $p = q = 1/2$, apply OST to Y_n itself:

$$0 = \mathbb{E}[Y_\tau] = \pi b - (1-\pi)a,$$

and hence $\pi = a/(a+b)$. □

The quadratic martingale gives more than the hitting side. In the symmetric case, bounded stopping applied to $Y_n^2 - n$ gives

$$\mathbb{E}[\tau \wedge N] = \mathbb{E}[Y_{\tau\wedge N}^2] \leq \max\{a^2, b^2\}.$$

Monotone convergence first proves $\mathbb{E}[\tau] < \infty$; bounded convergence for $Y^2_{\tau\wedge N}$ then yields

$$0 = \mathbb{E}[Y_\tau^2 - \tau] = \pi b^2 + (1-\pi)a^2 - \mathbb{E}[\tau], \tag{2.29}$$

$$\boxed{\mathbb{E}_0[\tau] = ab.} \tag{2.30}$$

Thus the same Rademacher increment produces both the harmonic exit probability and the mean absorption time.

To hold physical barriers $-A$ and $+B$ fixed, take

$$a = \frac{A}{\sigma\sqrt{\delta}}, \qquad b = \frac{B}{\sigma\sqrt{\delta}},$$

along a lattice-compatible sequence. Writing $\varepsilon = (\mu/\sigma)\sqrt{\delta}$,

$$\log \frac{q_\delta}{p_\delta} = \log \frac{1-\varepsilon}{1+\varepsilon} = -2\varepsilon + O(\varepsilon^3). \tag{2.31}$$

Hence

$$\left(\frac{q_\delta}{p_\delta}\right)^a \longrightarrow e^{-2\mu A/\sigma^2}, \qquad \left(\frac{q_\delta}{p_\delta}\right)^{a+b} \longrightarrow e^{-2\mu(A+B)/\sigma^2}. \tag{2.32}$$

Substitution in (2.28) gives the drifted Brownian exit formula derived again in Section 6 by optional stopping.

3 Gaussian MGF and the substitution $k = \theta\sigma$

3.1 Direct completion-of-the-square calculation

Proposition 3.1 (MGF of a Brownian marginal)**.** *For every $t > 0$ and $k \in \mathbb{R}$,*

$$\mathbb{E}[e^{kB_t}] = e^{k^2 t/2}. \tag{3.1}$$

Proof. The density of B_t is

$$p_t(x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right).$$

Therefore

$$\mathbb{E}[e^{kB_t}] = \frac{1}{\sqrt{2\pi t}} \int_{\mathbb{R}} \exp\left(kx - \frac{x^2}{2t}\right) \mathrm{d}x. \tag{3.2}$$

Complete the square:

$$kx - \frac{x^2}{2t} = -\frac{1}{2t}(x^2 - 2tkx) \tag{3.3}$$

$$= -\frac{(x-tk)^2}{2t} + \frac{k^2 t}{2}. \tag{3.4}$$

Substituting (3.4) into (3.2) gives

$$\mathbb{E}[e^{kB_t}] = e^{k^2 t/2} \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(x-tk)^2}{2t}\right) \mathrm{d}x \tag{3.5}$$

$$= e^{k^2 t/2}. \tag{3.6}$$

The remaining integrand is the density of $N(tk, t)$, so its integral is one. □

Corollary 3.2 (General normal MGF)**.** *If $Y \sim N(m, v)$, where $v \geq 0$, then*

$$M_Y(k) = \mathbb{E}[e^{kY}] = \exp\left(mk + \frac{1}{2}vk^2\right). \tag{3.7}$$

In particular, $m = 0$ and $v = t$ recover (3.1).

3.2 Why introduce $k = \theta\sigma$?

Insert (1.1) into the exponential:

$$e^{\theta X_t} = e^{\theta x_0} e^{\theta\mu t} e^{(\theta\sigma)B_t}, \tag{3.8}$$

$$\mathbb{E}[e^{\theta X_t}] = e^{\theta x_0} e^{\theta\mu t} \mathbb{E}[e^{(\theta\sigma)B_t}]. \tag{3.9}$$

The standard MGF formula is written for an exponent kB_t. The actual coefficient multiplying B_t in (3.9) is $\theta\sigma$. Thus

$$\boxed{k = \theta\sigma} \tag{3.10}$$

is exactly the coefficient-matching substitution needed to invoke (3.1). It is an abbreviation, not a new probabilistic assumption and not a change of measure.

Using (3.1),

$$\mathbb{E}[e^{\theta X_t}] = e^{\theta x_0} e^{\theta\mu t} e^{(\theta\sigma)^2 t/2} \tag{3.11}$$

$$= \exp\left[\theta x_0 + \left(\mu\theta + \frac{1}{2}\sigma^2\theta^2\right) t\right]. \tag{3.12}$$

Interpretation of the substitution

The volatility σ rescales the random noise before the exponential is taken. After coefficient matching, the Gaussian MGF turns this random term into the deterministic growth contribution

$$\frac{1}{2}(\theta\sigma)^2 t = \frac{1}{2}\theta^2\sigma^2 t.$$

This is the exact contribution that must be balanced against the drift term $\mu\theta t$.

4 Exponential-martingale calibration

Define the cumulant rate

$$\psi(\theta) := \mu\theta + \frac{1}{2}\sigma^2\theta^2. \tag{4.1}$$

Theorem 4.1 (Compensated exponential martingale). *For every $\theta \in \mathbb{R}$, the process*

$$\mathcal{E}_t(\theta) := \exp(\theta(X_t - x_0) - \psi(\theta)t) \tag{4.2}$$

is a positive, mean-one $(\mathcal{F}_t)$-martingale.

Proof. Let $0 \le s \le t$. From independent Brownian increments, $B_t - B_s \sim N(0, t-s)$ and is independent of $\mathcal{F}_s$. Dividing $\mathcal{E}_t(\theta)$ by $\mathcal{E}_s(\theta)$ yields

$$\frac{\mathcal{E}_t(\theta)}{\mathcal{E}_s(\theta)} = \exp\left(\theta\sigma(B_t - B_s) - \frac{1}{2}\theta^2\sigma^2(t-s)\right).$$

Conditional expectation and (3.1) give

$$\mathbb{E}[\mathcal{E}_t(\theta) \mid \mathcal{F}_s] = \mathcal{E}_s(\theta) e^{-\theta^2\sigma^2(t-s)/2} \mathbb{E}[e^{\theta\sigma(B_t - B_s)}] \tag{4.3}$$

$$= \mathcal{E}_s(\theta). \tag{4.4}$$

Since $\mathcal{E}_0(\theta) = 1$, its expectation is one at every time. □

Corollary 4.2 (Uncorrected calibration). *The process $e^{\theta(X_t - x_0)}$ is a martingale precisely when*

$$\psi(\theta) = 0. \tag{4.5}$$

The two roots are

$$\theta = 0 \qquad \text{and} \qquad \boxed{\theta_* = -\frac{2\mu}{\sigma^2}}. \tag{4.6}$$

For $x_0 = 0$, the nontrivial process $e^{\theta_* X_t}$ is mean one. For general x_0, the mean-one normalization is $e^{\theta_*(X_t - x_0)}$.

Proof. Ito's formula provides a local differential verification:

$$\mathrm{d}e^{\theta(X_t - x_0)} = e^{\theta(X_t - x_0)} \left[\theta\sigma\,\mathrm{d}B_t + \left(\mu\theta + \frac{1}{2}\sigma^2\theta^2\right)\mathrm{d}t\right]. \tag{4.7}$$

The finite-variation term vanishes exactly when (4.5) holds. The conditional-expectation argument above then upgrades the resulting local martingale to a true martingale. □

Remark 4.3 (Mean one is not, by itself, a martingale proof)**.** For an arbitrary adapted process, the condition $\mathbb{E}[M_t] = 1$ for every t does not imply $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$. Here the martingale property follows from independent increments, or equivalently from the stochastic-exponential argument; the constant expectation is a consequence.

5 Characteristic function and Brownian transition density

The Gaussian MGF is finite for every real argument and extends to complex arguments. Setting $k = i\omega$ in (3.1) gives

$$\phi_{B_t}(\omega) = \mathbb{E}[e^{i\omega B_t}] = e^{-t\omega^2/2}. \tag{5.1}$$

For the drifted process,

$$\phi_{X_t}(\omega) = \exp\left(i\omega(x_0 + \mu t) - \frac{1}{2}\sigma^2 t\omega^2\right). \tag{5.2}$$

5.1 Fourier inversion

For standard Brownian motion, (1.2) and (5.1) give

$$p_t(x) = \frac{1}{2\pi} I_t(x), \qquad I_t(x) = \int_{\mathbb{R}} e^{-t\omega^2/2 - ix\omega}\, \mathrm{d}\omega. \tag{5.3}$$

Complete the square:

$$-\frac{t}{2}\omega^2 - ix\omega = -\frac{t}{2}\left(\omega + \frac{ix}{t}\right)^2 - \frac{x^2}{2t}. \tag{5.4}$$

Thus

$$I_t(x) = e^{-x^2/(2t)} \int_{\mathbb{R}} e^{-\frac{t}{2}(\omega + ix/t)^2}\, \mathrm{d}\omega. \tag{5.5}$$

5.2 The entire-function contour shift

Lemma 5.1 (Horizontal contour-shift identity)**.** *For every $t > 0$ and every $h \in \mathbb{R}$,*

$$\int_{\mathbb{R}} e^{-t(u+ih)^2/2}\, \mathrm{d}u = \int_{\mathbb{R}} e^{-tu^2/2}\, \mathrm{d}u. \tag{5.6}$$

Proof. Let $g_t(z) = e^{-tz^2/2}$, which is entire. First assume $h > 0$ and use the counterclockwise rectangle with vertices $-R, R, R + ih, -R + ih$. The residue theorem gives

$$\oint_{\Gamma_R} g_t(z)\, \mathrm{d}z = 2\pi i \sum_{z_k \in \operatorname{int} \Gamma_R} \operatorname{Res}(g_t; z_k) = 0, \tag{5.7}$$

because there are no singularities.

On the right edge $z = R + iy$, $0 \le y \le h$,

$$|g_t(R + iy)| = \exp\left[-\frac{t}{2}(R^2 - y^2)\right] \le \exp\left(-\frac{tR^2}{2} + \frac{th^2}{2}\right).$$

The ML estimate therefore gives

$$\left|\int_{V_R} g_t(z)\, \mathrm{d}z\right| \le h \exp\left(-\frac{tR^2}{2} + \frac{th^2}{2}\right) \longrightarrow 0. \tag{5.8}$$

The left edge obeys the same bound. Letting $R \to \infty$ in (5.7) leaves the bottom and oppositely oriented top edges, which proves (5.6). For $h < 0$, use the rectangle below the real axis and replace h by $|h|$ in the estimate. The case $h = 0$ is immediate. □

Remark 5.2 (What the residue theorem does and does not compute)**.** The contour argument is valid, but all residues are zero. Its purpose is to justify moving the line of integration from $\operatorname{Im} z = 0$ to $\operatorname{Im} z = x/t$. It does not determine the Gaussian normalization constant. Also,

$$|e^{-tz^2/2}| = e^{-t((\operatorname{Re} z)^2 - (\operatorname{Im} z)^2)/2},$$

so the integrand grows along the imaginary axis. The vertical-edge argument works because the height $|h|$ is fixed while $R \to \infty$.

5.3 Gaussian normalization and the heat kernel

Proposition 5.3 (Gaussian integral). *For $t > 0$,*

$$\int_{\mathbb{R}} e^{-tu^2/2}\,\mathrm{d}u = \sqrt{\frac{2\pi}{t}}. \tag{5.9}$$

Proof. Scaling reduces the problem to $G = \int_{\mathbb{R}} e^{-u^2/2}\,\mathrm{d}u$. By Tonelli's theorem and polar coordinates,

$$G^2 = \iint_{\mathbb{R}^2} e^{-(u^2+v^2)/2}\,\mathrm{d}u\,\mathrm{d}v = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2} r\,\mathrm{d}r\,\mathrm{d}\varphi = 2\pi.$$

Since $G > 0$, $G = \sqrt{2\pi}$, and rescaling gives (5.9). □

Taking $h = x/t$ in (5.6), then using (5.5) and (5.9), gives

$$I_t(x) = e^{-x^2/(2t)}\sqrt{\frac{2\pi}{t}}.$$

Consequently,

Theorem 5.4 (Brownian heat kernel). *For each fixed $t > 0$, the exact marginal density of B_t is*

$$\boxed{p_t(x) = \frac{1}{\sqrt{2\pi t}}\exp\left(-\frac{x^2}{2t}\right)}. \tag{5.10}$$

It satisfies

$$\partial_t p_t = \frac{1}{2}\partial_{xx} p_t, \qquad \int_{\mathbb{R}} p_t(x)\,\mathrm{d}x = 1, \qquad p_t \longrightarrow \delta_0 \quad (t \downarrow 0)$$

in the distributional sense.

For $X_t = x_0 + \mu t + \sigma B_t$, the corresponding density is

$$p_X(x,t) = \frac{1}{\sigma\sqrt{2\pi t}}\exp\left[-\frac{(x - x_0 - \mu t)^2}{2\sigma^2 t}\right], \tag{5.11}$$

and its forward equation is

$$\partial_t p = -\mu\partial_x p + \frac{\sigma^2}{2}\partial_{xx} p.$$

6 Two-sided absorption via optional stopping

Assume $X_0 = 0$, fix $a, b > 0$, and define

$$\tau = \inf\{t \geq 0 : X_t \notin (-a, b)\}. \tag{6.1}$$

Continuity implies $X_\tau \in \{-a, b\}$, and exit from a bounded interval occurs almost surely.

Theorem 6.1 (Upper-boundary hitting probability). *Let*

$$p_+(\mu) = \mathbb{P}_0(X_\tau = b).$$

For $\mu \neq 0$,

$$\boxed{p_+(\mu) = \frac{1 - e^{-2\mu a/\sigma^2}}{1 - e^{-2\mu(a+b)/\sigma^2}}}. \tag{6.2}$$

At $\mu = 0$, its continuous extension is

$$p_+(0) = \frac{a}{a+b}. \tag{6.3}$$

Proof. Suppose $\mu \neq 0$ and set $\theta_* = -2\mu/\sigma^2$. Then $Y_t = e^{\theta_* X_t}$ is a martingale. Because $X_{t\wedge\tau} \in [-a, b]$, the stopped family $Y_{t\wedge\tau}$ is bounded and hence uniformly integrable. Optional stopping followed by $t \to \infty$ gives

$$1 = \mathbb{E}[e^{\theta_* X_\tau}] \tag{6.4}$$

$$= p_+(\mu)e^{\theta_* b} + (1 - p_+(\mu))e^{-\theta_* a}. \tag{6.5}$$

Solving for $p_+(\mu)$ and substituting θ_* yields (6.2). If $\mu = 0$, apply optional stopping directly to the martingale $X_t = \sigma B_t$:

$$0 = \mathbb{E}[X_\tau] = p_+(0)b - (1 - p_+(0))a,$$

which gives (6.3). □

Thus positive drift increases the probability of reaching $+b$ first, while negative drift increases the probability of reaching $-a$ first. For $a = b$, both probabilities equal $1/2$ at $\mu = 0$. Any plot showing the opposite monotonic directions has either exchanged the curve labels or reversed the sign convention for drift.

7 Python visualizations of the discrete–continuous limit

Figure 1 begins with the random-walk skeleton. Every panel uses exact formulas or simulations from the definitions above; the purple palette is anchored at NYU Violet #57068C.

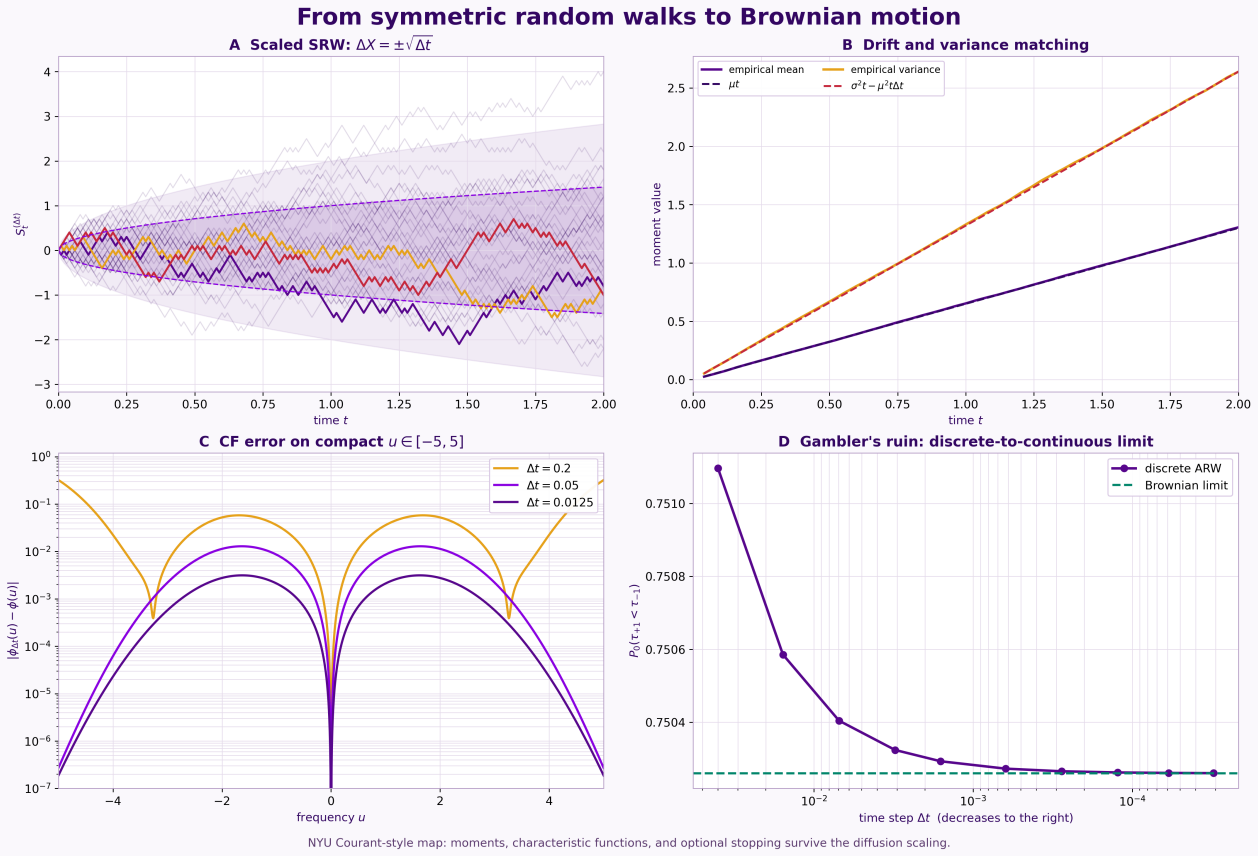


Figure 1: Diffusion scaling of SRW. The panels show scaled sample paths, the exact first two moments, locally uniform characteristic-function convergence on a fixed compact frequency interval, and convergence of discrete gambler's-ruin probabilities to the Brownian exit formula.

Panel	Object	Reading rule
A	Scaled paths	The spatial scale $\sigma\sqrt{\delta}$ produces fluctuations of order $\sqrt{t}$.
B	First two moments	The weak bias preserves $\mathbb{E}X_t = \mu t$ and sends $\mathrm{Var}(X_t)$ to $\sigma^2 t$.
C	Characteristic functions	On every fixed frequency window, the discrete CF approaches the Gaussian CF uniformly.
D	Ruin probability	The exact lattice exit law approaches the drifted Brownian exit formula.

Brownian, Fourier, and contour geometry

Figure 2 continues the same argument after the limiting Brownian process has been identified.

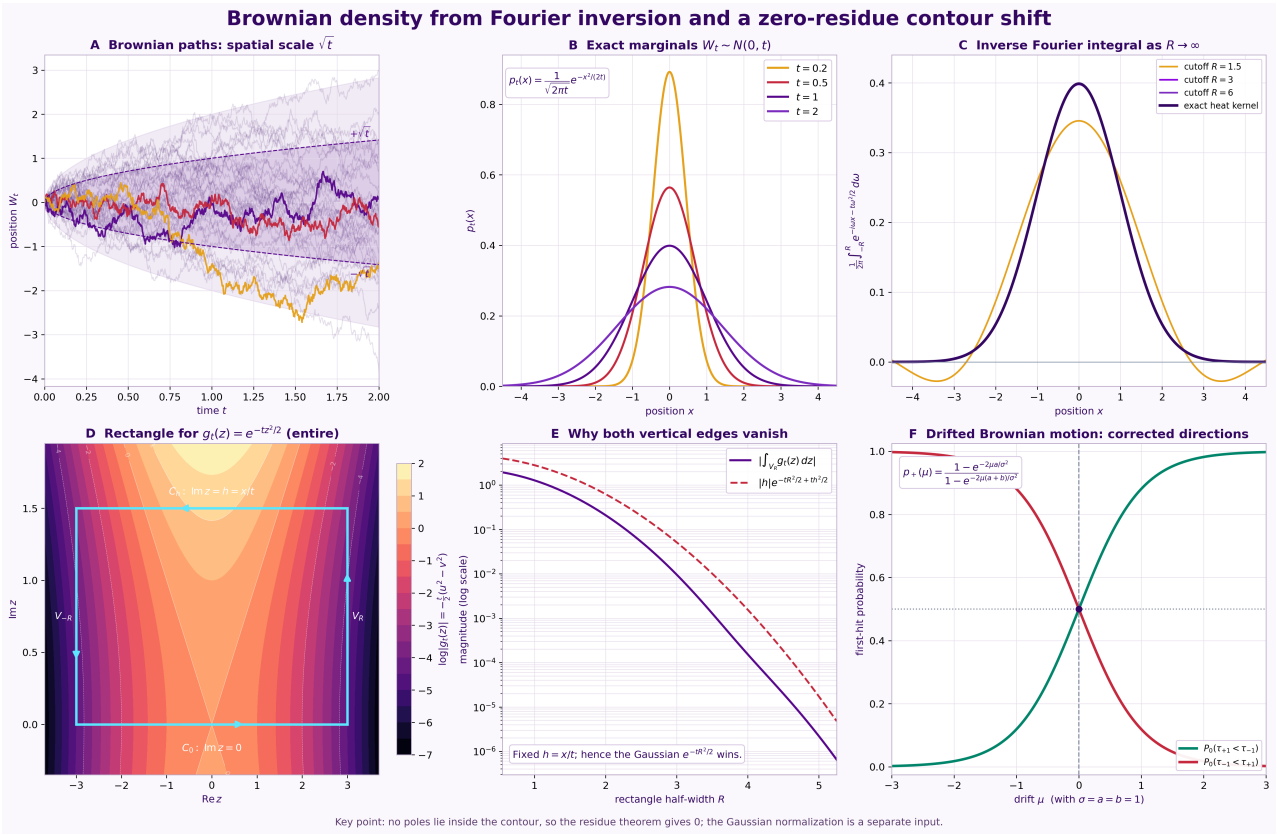


Figure 2: Brownian diffusion, Fourier inversion, the zero-residue contour shift, vertical-edge decay, and correctly oriented two-sided hitting probabilities for $a = b = \sigma = 1$.

Panel	Object	Reading rule
A	Sample paths	Typical displacement has order $\sqrt{t}$.
B	Heat kernels	At fixed time, $B_t \sim N(0,t)$.
C	Fourier cutoff	Truncated inversion converges to the heat kernel.
D	Rectangle	Shift the real axis to $\operatorname{Im} z = x/t$.
E	Vertical edges	The Gaussian estimate (5.8) makes both side integrals vanish.
F	Exit probabilities	Positive drift raises the probability of first hitting the upper barrier.

Transforms, devil's staircases, and tightness

Figure 3 separates a CDF jump caused by an atom from a characteristic-function discontinuity caused by escaped mass.

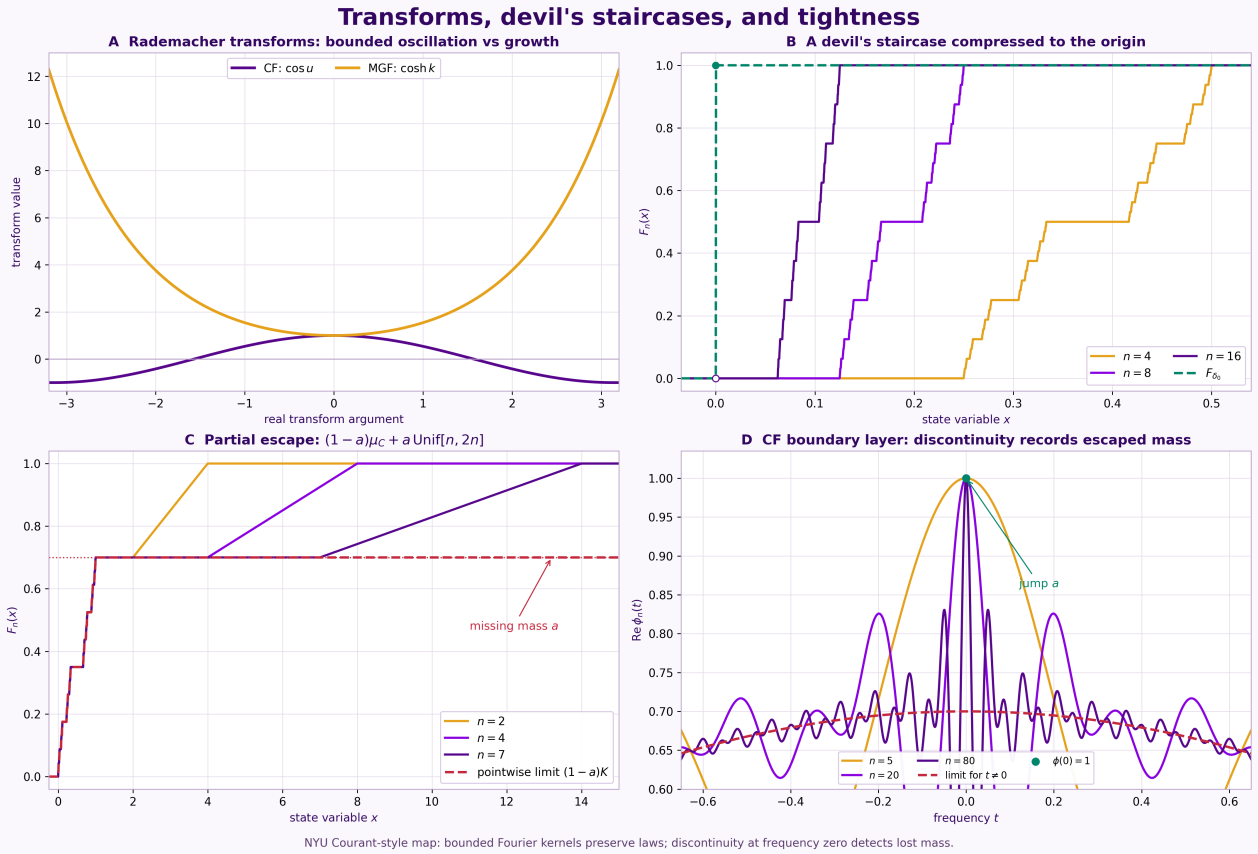


Figure 3: Rademacher CF versus MGF; Cantor staircases compressed toward δ_0; a Cantor component mixed with mass escaping through $[n, 2n]$; and the shrinking frequency boundary layer whose limiting jump at zero equals the escaped mass.

Panel	Object	Diagnostic meaning
A	CF versus MGF	Oscillation keeps the CF bounded; real exponential weighting can make the MGF diverge.
B	Compressed Cantor CDFs	Weak convergence need not hold at an atom of the limiting CDF.
C	Escaping mixture	A fixed proportion of probability mass leaves every compact state-space set.
D	Frequency boundary layer	The limiting CF jump at zero records exactly the escaped mass.

8 Modes of convergence

8.1 Pointwise, uniform, almost-everywhere, and in-measure convergence

Let $(E, \mathcal{A}, \nu)$ be a measure space and let $f_n, f : E \to \mathbb{R}$ be measurable.

Definition 8.1 (Pointwise convergence)**.** $f_n \to f$ pointwise on E when

$$\forall x \in E,\ \forall \varepsilon > 0,\ \exists N = N(x, \varepsilon) \text{ such that } n \geq N \Longrightarrow |f_n(x) - f(x)| < \varepsilon.$$

The index N may depend on the point x.

Definition 8.2 (Uniform convergence)**.** $f_n \to f$ uniformly on E when

$$\forall \varepsilon > 0,\ \exists N = N(\varepsilon) \text{ such that } n \geq N \Longrightarrow \sup_{x \in E} |f_n(x) - f(x)| < \varepsilon.$$

One index N must work simultaneously for every x.

Definition 8.3 (Almost-everywhere and in-measure convergence)**.** We write $f_n \to f$ almost everywhere if pointwise convergence fails only on a ν-null set. We write $f_n \to f$ in measure when, for every $\varepsilon > 0$,

$$\nu(\{x : |f_n(x) - f(x)| > \varepsilon\}) \longrightarrow 0.$$

Proposition 8.4 (Basic implication chain on a finite measure space)**.** *If $\nu(E) < \infty$, then*

$$\textit{uniform} \Longrightarrow \textit{pointwise} \Longrightarrow \textit{almost everywhere} \Longrightarrow \textit{in measure}.$$

The converses fail in general.

Proof. Only the final implication needs comment. If $f_n \to f$ almost everywhere, then for each $\varepsilon > 0$ the indicators

$$\mathbf{1}_{\{|f_n - f| > \varepsilon\}} \longrightarrow 0 \quad \text{almost everywhere.}$$

They are bounded by the integrable function 1 because $\nu(E) < \infty$. Dominated convergence therefore makes their integrals, which are exactly the measures of the bad sets, tend to zero. □

Remark 8.5 (Probability language)**.** On a probability space, convergence in measure is precisely convergence in probability. For random variables,

$$X_n \xrightarrow{\text{a.s.}} X \Longrightarrow X_n \xrightarrow{\mathbb{P}} X \Longrightarrow X_n \Rightarrow X.$$

Weak convergence $X_n \Rightarrow X$ is a statement about laws and is not pointwise convergence of sample paths.

8.2 Sharp counterexamples and the Riesz subsequence theorem

For $f_n(x) = x^n$ on $[0, 1]$,

$$f_n(x) \longrightarrow f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

This convergence is pointwise but not uniform: for every n,

$$\sup_{0 \leq x \leq 1} |x^n - f(x)| = 1.$$

The discontinuous limit is also consistent with the theorem that a uniform limit of continuous functions is continuous.

For the reverse failure, use the typewriter sequence on the unit circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. If $m \geq 0$ and $0 \leq k < 2^m$, set

$$f_{2^m + k} = \mathbf{1}_{[k2^{-m}, (k+1)2^{-m})}. \tag{8.1}$$

The support has Lebesgue measure 2^{-m}, so $f_n \to 0$ in measure. At each point, however, the sequence takes the value 1 once in every dyadic block and 0 at all other positions in that block. It therefore has no pointwise limit anywhere on $\mathbb{T}$.

Theorem 8.6 (Riesz subsequence principle)**.** *If $f_n \to f$ in measure, then there exists a subsequence (f_{n_j}) such that $f_{n_j} \to f$ almost everywhere.*

Proof idea. Choose n_j so that

$$\nu\big(|f_{n_j} - f| > 2^{-j}\big) < 2^{-j}.$$

The sum of these measures is finite. The Borel–Cantelli lemma implies that almost every point belongs to only finitely many of the bad sets, which is exactly almost-everywhere convergence of the chosen subsequence. □

Table 2: Three deterministic modes of convergence.

Mode	Quantitative test	Dependence of N	What it controls
Uniform	$\|f_n - f\|_\infty \to 0$	$N = N(\varepsilon)$, independent of x	Worst-case vertical error over the entire domain.
Pointwise	$f_n(x) \to f(x)$ for every fixed x	$N = N(x, \varepsilon)$	One scalar sequence at a time; convergence rates may vary arbitrarily with x.
In measure	$\nu(\|f_n - f\| > \varepsilon) \to 0$	No pointwise index is controlled	The size of the bad set; individual points may keep oscillating.

8.3 Equicontinuity and the Arzelà–Ascoli bridge

A family $\mathcal{G} \subset C(K)$ on a metric space K is *equicontinuous* if

$$\forall \varepsilon > 0,\ \exists \delta > 0 \quad d(x, y) < \delta \Longrightarrow |g(x) - g(y)| < \varepsilon \quad \text{for every } g \in \mathcal{G}.$$

Proposition 8.7 (Two useful upgrades on a compact set)**.** *Let K be compact.*

(i) If $f_n \in C(K)$ and $f_n \to f$ uniformly, then $f \in C(K)$ and the family $\{f_n : n \geq 1\}$ is equicontinuous.
(ii) If $\{f_n\}$ is equicontinuous and $f_n \to f$ pointwise on K, then $f_n \to f$ uniformly.

Why compactness closes the gaps. For (i), uniform convergence transfers continuity to f. The uniformly close tail is controlled by the uniform continuity of f, while the finitely many initial functions have a common modulus after taking the minimum of finitely many admissible radii. For (ii), equicontinuity first passes to the pointwise limit f. A finite δ-net of K then reduces the required estimate to pointwise convergence at finitely many centers. □

Theorem 8.8 (Arzelà–Ascoli)**.** *For compact metric K, a subset $\mathcal{G} \subset C(K)$ is relatively compact in the uniform norm if and only if it is pointwise bounded and equicontinuous. Equivalently, every sequence in such a family contains a uniformly convergent subsequence.*

Remark 8.9 (What uniform convergence does not preserve)**.** Uniform limits preserve continuity, and on a finite measure space satisfy

$$\left| \int_E f_n \, \mathrm{d}\nu - \int_E f \, \mathrm{d}\nu \right| \leq \nu(E) \|f_n - f\|_\infty.$$

They do not automatically preserve differentiability. For example, $f_n(x) = \sqrt{x^2 + 1/n}$ is differentiable and converges uniformly on $[-1, 1]$ to $|x|$, which is not differentiable at zero. A standard sufficient derivative theorem instead assumes uniform convergence of f_n' together with convergence at one anchor point.

9 Lévy continuity and the passage from laws to paths

9.1 The exact continuity theorem

Theorem 9.1 (Lévy's continuity theorem)**.** *Let ϕ_n be the characteristic function of X_n.*

(i) If $X_n \Rightarrow X$, then

$$\phi_n(u) \longrightarrow \phi_X(u) \qquad \text{for every } u \in \mathbb{R},$$

and the convergence is uniform on each compact u-set.
(ii) Conversely, if $\phi_n(u) \to \phi(u)$ pointwise and ϕ is continuous at zero, then ϕ is the characteristic function of a probability law and X_n converges in distribution to that law.

The word pointwise refers to frequency

The hypothesis is pointwise convergence of the deterministic functions $u \mapsto \phi_n(u)$. It is not convergence in measure of X_n, nor pointwise convergence of the random variables $\omega \mapsto X_n(\omega)$. The conclusion is weak convergence of probability laws.

Global uniform convergence on all of $\mathbb{R}$ is false in general. Indeed, let $X_n \equiv 1/n$ and $X \equiv 0$. Then $X_n \Rightarrow X$ and $\phi_n(u) = e^{iu/n} \to 1$ pointwise, but

$$\sup_{u\in\mathbb{R}} |e^{iu/n} - 1| = 2 \qquad \text{for every } n. \tag{9.1}$$

Thus any unqualified claim of uniformity must here be read as *locally uniform*: uniform on each fixed compact frequency interval.

9.2 A compressed Cantor staircase: a CDF jump is not a CF discontinuity

Let $(A_k)_{k\geq 1}$ be independent Bernoulli(1/2) variables and define

$$C = \sum_{k=1}^{\infty} \frac{2A_k}{3^k}, \qquad K(x) = \mathbb{P}(C \leq x). \tag{9.2}$$

The law of C is the standard Cantor distribution and K is the Cantor–Lebesgue devil's staircase: it is continuous and nondecreasing, and is constant on every removed middle-third interval.

Now set

$$X_n = \frac{1 + C}{n}. \tag{9.3}$$

Its support is $[1/n, 2/n]$, and its CDF is

$$F_n(x) = K(nx - 1).$$

For every fixed x,

$$F_n(x) \longrightarrow G(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases} \tag{9.4}$$

The pointwise limit G is not right-continuous at zero, so it is not a CDF. Nevertheless $0 \leq X_n \leq 2/n$, hence

$$X_n \longrightarrow 0 \quad \text{almost surely and therefore in distribution.}$$

The actual weak limit is δ_0, whose right-continuous CDF is

$$F(x) = \mathbf{1}_{[0,\infty)}(x).$$

At the atom $x = 0$, $F_n(0) = 0$ while $F(0) = 1$. This does not contradict the CDF criterion for weak convergence, which requires $F_n(x) \to F(x)$ only at continuity points of F. In fact,

$$\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| = 1. \tag{9.5}$$

This sequence is maximally tight for the present purpose: all its laws are supported in the single compact interval $[0, 2]$. Tightness prevents mass from escaping, but it does not force convergence of CDFs at atoms.

The characteristic functions have no defect:

$$\phi_{X_n}(t) = e^{it/n}\phi_C(t/n) \longrightarrow 1. \tag{9.6}$$

Moreover, for every fixed $T < \infty$,

$$\sup_{|t|\leq T} |\phi_{X_n}(t) - 1| \leq \sup_{|t|\leq T} \mathbb{E}|e^{itX_n} - 1| \tag{9.7}$$

$$\leq \frac{2T}{n} \longrightarrow 0. \tag{9.8}$$

Thus a jump of the *limiting CDF in the state variable* x is entirely compatible with a characteristic-function limit continuous at the *frequency origin* $t = 0$.

Theorem 9.2 (Pólya's uniform CDF theorem)**.** *If F_n converges weakly to a continuous CDF F, then*

$$\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| \longrightarrow 0.$$

Consequently, the obstruction in (9.5) is the atom of δ_0, not a failure of tightness. For example, if $Z \sim N(0, 1)$ and Z is added to every X_n, then $Z + X_n \Rightarrow Z$; the limiting CDF is continuous, and Pólya's theorem upgrades the convergence to uniform convergence on the whole real line.

9.3 A partially escaping staircase: discontinuity at frequency zero

Fix $0 < a < 1$ and let $U_n \sim \mathrm{Unif}[n, 2n]$. Consider the mixture

$$\mu_n = (1-a)\mu_C + a\mu_{U_n}. \tag{9.9}$$

Its CDF is

$$F_n(x) = (1-a)K(x) + aH_n(x), \qquad H_n(x) = \begin{cases} 0, & x < n, \\ (x-n)/n, & n \le x \le 2n, \\ 1, & x > 2n. \end{cases} \tag{9.10}$$

For each fixed finite x, eventually $x < n$, and therefore

$$F_n(x) \longrightarrow G(x) := (1-a)K(x). \tag{9.11}$$

The Cantor staircase remains, but its terminal height is only $1-a$. Thus G is a subprobability CDF: a mass a has escaped to $+\infty$.

The characteristic function makes exactly the same loss visible. Write

$$\phi_n(t) = (1-a)\phi_C(t) + a\psi_n(t),$$

where

$$\psi_n(t) = \begin{cases} \dfrac{e^{2int} - e^{int}}{int}, & t \neq 0, \\ 1, & t = 0. \end{cases} \tag{9.12}$$

For fixed $t \neq 0$,

$$|\psi_n(t)| \le \frac{2}{n|t|} \longrightarrow 0.$$

Hence the pointwise limit is

$$\phi(t) = \begin{cases} 1, & t = 0, \\ (1-a)\phi_C(t), & t \neq 0. \end{cases} \tag{9.13}$$

Since $\phi_C(t) \to 1$ as $t \to 0$,

$$\phi(0) - \lim_{\substack{t \to 0 \\ t \neq 0}} \phi(t) = a. \tag{9.14}$$

The missing terminal CDF height, the escaped probability mass, and the jump of the pointwise CF limit are the same number a.

The family (μ_n) is not tight. Given a fixed M, once $n > M$, the entire uniform component lies outside $[-M, M]$, so

$$\mu_n([-M, M]) \le 1 - a.$$

For any $\varepsilon < a$, no single compact interval can contain probability at least $1 - \varepsilon$ for all n. On the extended line one may symbolically write

$$\mu_n \Rightarrow (1-a)\mu_C + a\delta_{+\infty},$$

but on $\mathbb{R}$ the sequence has no probability weak limit.

The failure of uniform convergence is sharply localized near zero. For every $\delta > 0$,

$$\sup_{|t| \ge \delta} |\phi_n(t) - \phi(t)| \le \frac{2a}{n\delta} \longrightarrow 0. \tag{9.15}$$

On any compact interval containing zero, however, uniform convergence is impossible: every ϕ_n is continuous while the pointwise limit (9.13) is not. The defect is a boundary layer of frequency width comparable to $1/n$ collapsing onto the origin.

Example 9.3 (The visible boundary layer at $a = 0.35$)**.** Take $a = 0.35$, so the fixed Cantor component has mass $1 - a = 0.65$. For every finite state-space point,

$$F_n(x) \longrightarrow 0.65K(x), \qquad \lim_{x \to \infty} 0.65K(x) = 0.65.$$

Thus the staircase deficit is 0.35. On the original frequency scale the uniform component becomes invisible at every fixed $t \neq 0$, but the rescaled coordinate $s = nt$ resolves the collapsing layer:

$$\phi_n(s/n) = 0.65\phi_C(s/n) + 0.35\frac{e^{2is} - e^{is}}{is}, \quad \phi_n(s/n) \longrightarrow \Phi_{\mathrm{BL}}(s) := 0.65 + 0.35\frac{e^{2is} - e^{is}}{is}, \tag{9.16}$$

where the quotient is defined continuously as 1 at $s = 0$. Consequently $\Phi_{\mathrm{BL}}(0) = 1$, whereas $\Phi_{\mathrm{BL}}(s) \to 0.65$ as $|s| \to \infty$. The values $n = 1, 6, 40$ in Figure 4 show the same conserved defect on the CDF scale, the fixed frequency scale, and the blown-up scale.

Two different origins

An atom of a weak limit may create a jump of its CDF at a state-space point $x = 0$; weak convergence simply does not test that discontinuity. A jump of the pointwise characteristic-function limit at the frequency origin $t = 0$ has a different meaning: mass has escaped and the limiting object is not a probability law on $\mathbb{R}$.

Table 3: The two devil's-staircase constructions isolate different failures.

Feature	Compressed staircase $X_n = (1 + C)/n$	Escaping mixture $(1 - a)\mu_C + a\mu_{U_n}$
Mass behavior	Concentrates at the finite point 0	A proportion a escapes to $+\infty$
Tightness	Yes; all supports lie in $[0, 2]$	No for every tolerance $\varepsilon < a$
Weak limit on $\mathbb{R}$	The probability law δ_0	No probability weak limit
CDF phenomenon	Pointwise limit chooses the wrong value at the atom $x = 0$	Pointwise limit has terminal height $1 - a$
CF phenomenon	$\phi_n \to 1$, locally uniformly and continuously at 0	Pointwise limit jumps by a at frequency 0

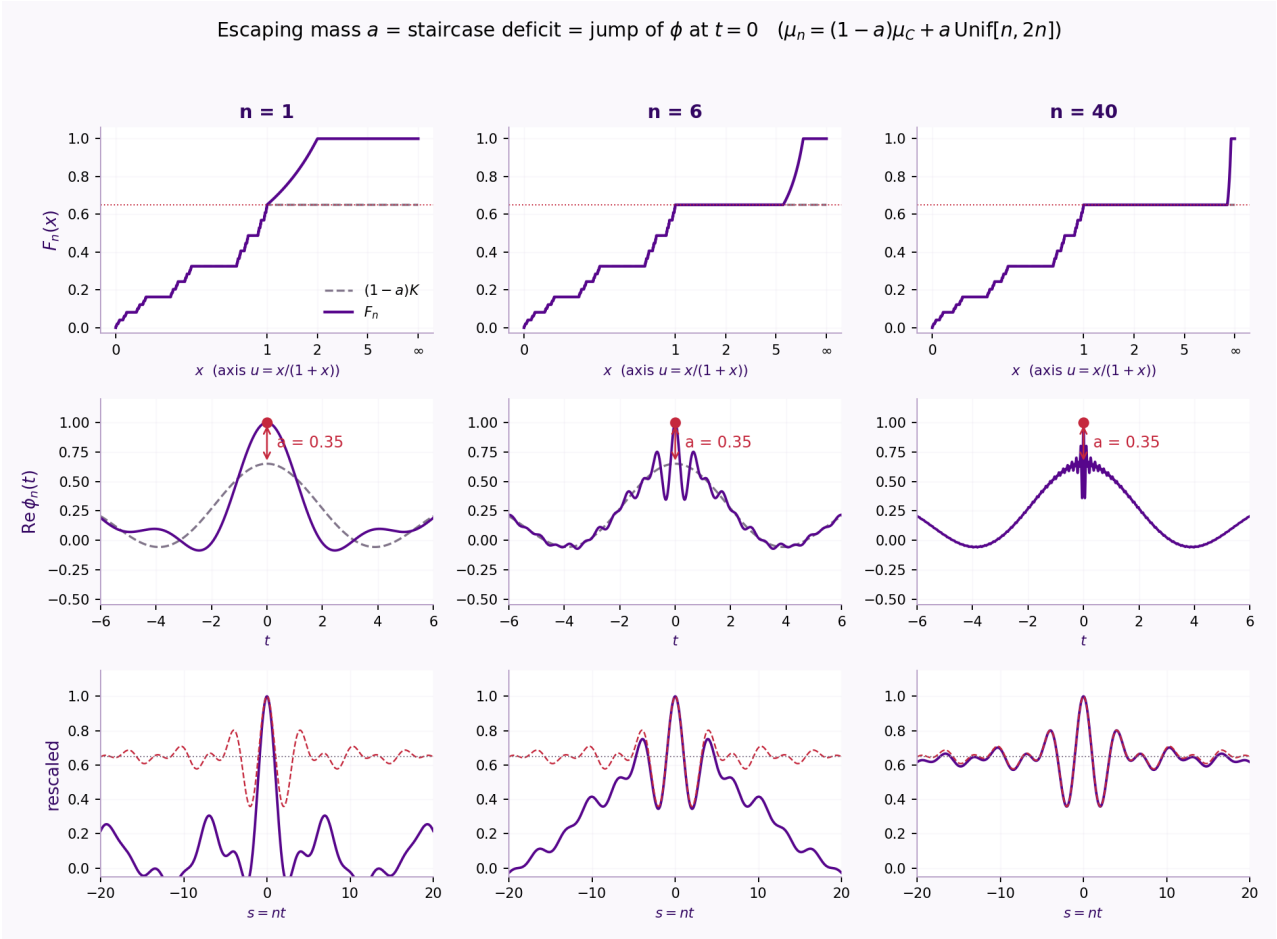


Figure 4: The escaping-mixture example with $a = 0.35$. Top: compactifying $[0, \infty]$ by $u = x/(1+x)$ keeps both the Cantor staircase and the moving uniform slab visible. Middle: the CF contribution concentrates at $t = 0$. Bottom: the rescaling $s = nt$ reveals the nontrivial boundary profile (9.16).

Scale	What the row isolates
State space x	The Cantor component stabilizes at height $1 - a$, while the uniform slab moves toward $+\infty$.
Fixed frequency t	For every $t \neq 0$, the escaping component vanishes, but it remains equal to a at $t = 0$.
Blown-up frequency $s = nt$	The shrinking layer becomes the nontrivial profile in (9.16).

9.4 Tightness, Prokhorov compactness, and local uniformity

A family of probability measures $\{\mu_n\}$ on $\mathbb{R}$ is tight when

$$\forall \varepsilon > 0, \quad \exists M < \infty, \quad \inf_n \mu_n([-M, M]) \geq 1 - \varepsilon. \tag{9.17}$$

Prokhorov's theorem identifies tightness with relative compactness in the weak topology: every sequence of laws then has a weakly convergent subsequence. In CDF language, Helly selection first supplies a monotone subsequential limit; tightness supplies the missing endpoint conditions

$$F(-\infty) = 0, \qquad F(+\infty) = 1.$$

Tightness also gives a common modulus of continuity for characteristic functions. If $\phi_n(t) = \int e^{itx} \mu_n(\,\mathrm{d}x)$, then for every $M > 0$,

$$|\phi_n(t+h) - \phi_n(t)| \leq \int_{\mathbb{R}} |e^{ihx} - 1| \, \mu_n(\,\mathrm{d}x) \tag{9.18}$$

$$\leq M|h| + 2\mu_n(|x| > M). \tag{9.19}$$

First choose M uniformly by tightness and then choose $|h|$ small. Thus $\{\phi_n\}$ is equicontinuous. On a fixed compact frequency interval, a finite mesh and pointwise convergence then give uniform convergence:

$$\text{tight laws} \Longrightarrow \text{equicontinuous CFs} \xrightarrow{\text{finite cover of } [-T,T]} \sup_{|t| \leq T} |\phi_n(t) - \phi(t)| \to 0. \tag{9.20}$$

This is the compactness mechanism behind the locally uniform part of Lévy's theorem.

The quantifiers must remain in their proper order:

$$\forall T < \infty, \qquad \lim_{n \to \infty} \sup_{|t| \leq T} |\phi_n(t) - \phi(t)| = 0.$$

It says nothing about a frequency window moving to infinity with n. For $X_n \equiv 1/n$,

$$|\phi_{X_n}(\pi n) - 1| = |e^{i\pi} - 1| = 2,$$

even though the laws are tight and convergence is uniform on every preassigned compact interval.

Definition 9.4 (Pisot and Rajchman)**.** A *Pisot number* is a real algebraic integer $\theta > 1$ whose other algebraic conjugates have modulus strictly below one. A finite measure is *Rajchman* when its Fourier transform tends to zero at infinity.

Example 9.5 (Powers approaching integers)**.** Let $\|y\| := \min_{m \in \mathbb{Z}} |y - m|$. For the golden ratio $\varphi = (1 + \sqrt{5})/2$, its conjugate $\varphi' = (1 - \sqrt{5})/2$ satisfies

$$\varphi^m + (\varphi')^m \in \mathbb{Z}, \qquad \|\varphi^m\| \leq |\varphi'|^m \longrightarrow 0.$$

Likewise $1 + \sqrt{2}$ has conjugate $1 - \sqrt{2}$, and its powers approach integers exponentially. In contrast, the rational numbers $3/2$ and $5/2$ are not algebraic integers and hence are not Pisot; their exact remainders modulo one do not converge to zero. The first panel of Figure 5 computes these four examples without floating point exponentiation for the rational cases.

Proposition 9.6 (Pisot resonance for a Bernoulli convolution)**.** *Let $(\varepsilon_k)_{k \geq 0}$ be independent Rademacher variables and, for $0 < \rho < 1$, define*

$$Y_\rho = \sum_{k=0}^{\infty} \varepsilon_k \rho^k, \qquad \Phi_\rho(t) = \mathbb{E}[e^{itY_\rho}] = \prod_{k=0}^{\infty} \cos(\rho^k t).$$

If ρ^{-1} is Pisot, the factors become simultaneously close to ± 1 along suitable geometric frequencies, and Φ_ρ need not decay. For $\rho = 1/3$, the explicit sequence $t_m = 2\pi 3^m$ gives

$$|\Phi_{1/3}(t_m)| = \prod_{j=1}^{\infty} \left| \cos \frac{2\pi}{3^j} \right| > 0 \qquad \textit{for every } m. \tag{9.21}$$

By the Erdős–Salem criterion, if ρ^{-1} is not Pisot then this Bernoulli convolution is Rajchman. In particular,

$$\Phi_{2/5}(t) \longrightarrow 0 \qquad (|t| \to \infty).$$

The finite-range envelopes in Figure 5 illustrate the contrast; they are numerical diagnostics, while the asymptotic conclusion is the analytic theorem.

For $\rho = 2/5$, the law of Y_ρ is purely singular continuous because the two self-similar pieces are separated. Nevertheless, for $Z_n = \rho^{-n} Y_\rho$,

$$\mathbb{P}(|Z_n| \le M) = \mathbb{P}(|Y_\rho| \le M\rho^n) \longrightarrow 0 \qquad (M < \infty),$$

so the laws escape from every compact set while remaining singular continuous, and

$$\phi_{Z_n}(t) = \Phi_\rho(\rho^{-n} t) \longrightarrow \mathbf{1}_{\{0\}}(t).$$

Thus singularity of a measure and decay of its Fourier transform are logically independent issues.

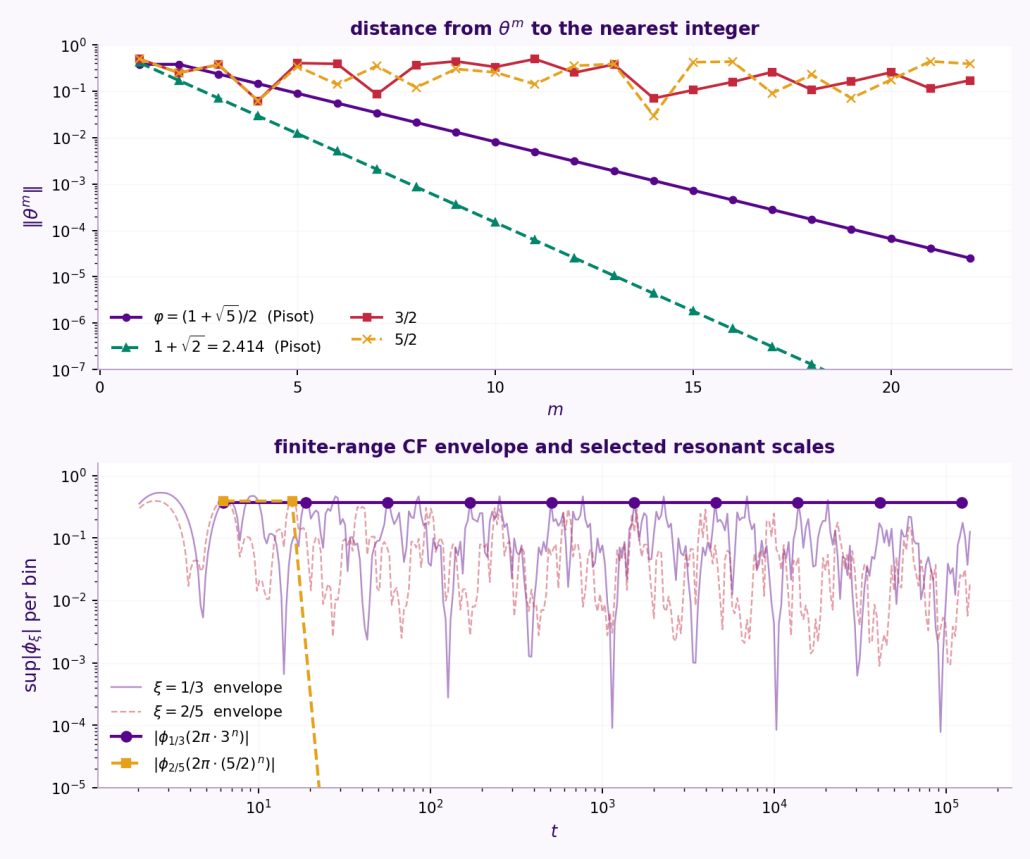


Figure 5: Pisot arithmetic and Fourier resonance. Top: powers of the quadratic Pisot examples approach integers exponentially, whereas the rational non-Pisot examples do not. Bottom: $\rho = 1/3$ retains a positive resonant subsequence (9.21); the $\rho = 2/5$ curve is a finite-range numerical view of the Rajchman case.

9.5 Weak convergence does not remember the coupling

Let

$$\Omega = \{-1, 1\}, \qquad \mathbb{P}(\{-1\}) = \mathbb{P}(\{1\}) = \frac{1}{2},$$

and define

$$X(\omega) = \omega, \qquad X_n(\omega) = -\omega + \frac{1}{n}. \tag{9.22}$$

Then X is Rademacher and

$$\mathcal{L}(X_n) = \frac{1}{2}\delta_{-1+1/n} + \frac{1}{2}\delta_{1+1/n} \Rightarrow \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1 = \mathcal{L}(X).$$

Indeed,

$$\phi_{X_n}(t) = e^{it/n}\cos t \longrightarrow \cos t = \phi_X(t). \tag{9.23}$$

The convergence is uniform on every fixed compact set, but at $t_n = \pi n$,

$$|\phi_{X_n}(t_n) - \phi_X(t_n)| = 2.$$

The CDFs also fail to converge uniformly:

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| = \frac{1}{2}, \tag{9.24}$$

because the limiting law has atoms at ± 1.

On the given probability space the variables do not converge in probability. For every $n \geq 2$,

$$\mathbb{P}(|X_n - X| > 1) = 1. \tag{9.25}$$

The marginal laws approach one another, while the coupling sends almost every sample point toward the opposite sign.

An even stronger version removes the $1/n$: let $Y_n = -X$ for every n. By Rademacher symmetry,

$$\phi_{Y_n}(t) = \phi_X(t) = \cos t \quad \text{for all } n, t,$$

so the characteristic functions agree globally, yet

$$\mathbb{P}(|Y_n - X| > 1) = 1.$$

Characteristic functions see probability laws; convergence in probability also sees how the variables are coupled on the same sample space.

9.6 Fixed-time convergence of the weakly biased walk

The one-step characteristic function of ΔX_j^δ is

$$\varphi_\delta(u) = p_\delta e^{iu\sigma\sqrt{\delta}} + q_\delta e^{-iu\sigma\sqrt{\delta}} \tag{9.26}$$

$$= \cos(u\sigma\sqrt{\delta}) + i\frac{\mu}{\sigma}\sqrt{\delta}\,\sin(u\sigma\sqrt{\delta}) \tag{9.27}$$

$$= 1 + \left(i\mu u - \frac{1}{2}\sigma^2 u^2\right)\delta + O(\delta^2), \tag{9.28}$$

where the error is uniform for u in any fixed compact set. If $N = t/\delta$, independence gives

$$\phi_{X_t^\delta}(u) = \big(\varphi_\delta(u)\big)^{t/\delta} \tag{9.29}$$

$$\longrightarrow \exp\!\left(i\mu u t - \frac{1}{2}\sigma^2 u^2 t\right). \tag{9.30}$$

The limit is continuous at zero and is the characteristic function of $N(\mu t, \sigma^2 t)$. Lévy's theorem therefore yields

$$X_t^\delta \Rightarrow \mu t + \sigma B_t \qquad \text{for every fixed } t \geq 0. \tag{9.31}$$

This proves convergence of one-time marginals, not yet convergence of whole sample paths.

9.7 Donsker's theorem supplies tightness

For the standard SRW, define the linearly interpolated process on $[0, T]$ by

$$W^{(n)}(t) = \frac{1}{\sqrt{n}}\left[S_{\lfloor nt\rfloor} + (nt - \lfloor nt\rfloor)\xi_{\lfloor nt\rfloor+1}\right]. \tag{9.32}$$

Theorem 9.7 (Donsker invariance principle)**.** *For every $T < \infty$,*

$$W^{(n)} \Rightarrow B \quad \text{in } C([0,T]) \text{ equipped with the uniform topology.}$$

For the triangular array with increments (2.16), the analogous interpolation satisfies

$$X^\delta \Rightarrow \big(t \mapsto \mu t + \sigma B_t\big) \quad \text{in } C([0,T]).$$

The logical distinction is important:

$$\underbrace{\text{Lévy continuity}}_{\text{one-time laws}} + \underbrace{\text{independent increments}}_{\text{finite-dimensional laws}} + \underbrace{\text{tightness}}_{\text{between-time control}} \Longrightarrow \underbrace{\text{Donsker path convergence}}_{\text{entire process}}.$$

The deterministic square-sum identity (2.21) explains the correct limiting variance clock, but tightness is the additional ingredient that rules out uncontrolled oscillation between observation times.

9.8 A second theorem bearing Lévy's name

Theorem 9.8 (Lévy's martingale characterization)**.** *If M is a continuous local martingale with $M_0 = 0$ and*

$$\langle M \rangle_t = t \qquad \text{for every } t \geq 0 \text{ almost surely,} \tag{9.33}$$

then M is standard Brownian motion relative to the given filtration.

The continuity theorem connects characteristic-function convergence with weak convergence of laws. The martingale characterization identifies Brownian motion from path continuity, the local-martingale property, and quadratic variation.

10 Measure-theoretic construction of Wiener measure

The finite-dimensional Gaussian formulas used above do not by themselves produce a random element of $C([0,T])$. There are two logically separate steps:

$$\text{consistent finite-dimensional laws} \xRightarrow{\text{Kolmogorov extension}} \text{a process on } \mathbb{R}^{[0,\infty)},$$

$$\text{a process on } \mathbb{R}^{[0,\infty)} \xRightarrow{\text{continuity criterion}} \text{a continuous modification.}$$

The first arrow is measure extension; the second is a path-regularity theorem.

10.1 Consistent Gaussian laws and the extension theorem

Theorem 10.1 (Kolmogorov extension theorem)**.** *Let I be an arbitrary index set and let E be a standard Borel space. For every finite ordered set $J = (t_1, \dots, t_m) \subset I$, suppose a probability measure μ_J on E^m is given. Assume permutation consistency and marginal consistency: relabeling coordinates relabels the measure, and deleting coordinates gives the corresponding lower-dimensional marginal. Then there is a unique probability measure $\mathbb{P}^0$ on*

$$\big(E^I,\ \sigma(\pi_t : t \in I)\big)$$

whose finite-dimensional marginals are the μ_J.

The proof first defines a finitely additive premeasure on cylinder sets by $\mathbb{P}^0(\pi_J^{-1}A) = \mu_J(A)$. Consistency makes this definition independent of the chosen presentation of a cylinder. Because E is standard Borel, tightness of each finite-dimensional probability law supplies the continuity from above needed for countable additivity on the cylinder algebra. Carathéodory extension then produces $\mathbb{P}^0$, and the cylinder sets form a generating π-system, so uniqueness follows from the π-λ theorem.

For times $t_1, \dots, t_m \geq 0$, define $\gamma_{t_1,\dots,t_m}$ to be the centered Gaussian law with covariance matrix

$$\Sigma_{ij} = t_i \wedge t_j. \tag{10.1}$$

Equivalently, its characteristic function is

$$\widehat{\gamma}_{t_1,\dots,t_m}(u) = \exp\left(-\frac{1}{2}\sum_{i,j=1}^{m} u_i u_j (t_i \wedge t_j)\right). \tag{10.2}$$

The matrix is positive semidefinite because

$$\sum_{i,j} u_i u_j (t_i \wedge t_j) = \int_0^\infty \left(\sum_i u_i \mathbf{1}_{[0,t_i]}(r)\right)^2 \mathrm{d}r \geq 0.$$

Setting a coordinate of u equal to zero in (10.2) deletes that coordinate, and permuting u and the times leaves the same quadratic form. Hence the family is consistent. Kolmogorov extension gives a probability measure on $\Omega_0 = \mathbb{R}^{[0,\infty)}$ and a coordinate process $B_t^0(\omega) = \omega(t)$ with these finite-dimensional laws.

Proposition 10.2 (Brownian increment structure before path regularization)**.** *The coordinate process satisfies $B_0^0 = 0$ almost surely, and for $0 \leq t_0 < t_1 < \cdots < t_m$, the increments*

$$B_{t_1}^0 - B_{t_0}^0, \dots, B_{t_m}^0 - B_{t_{m-1}}^0$$

are independent with respective laws $N(0, t_j - t_{j-1})$.

Proof. All the increments are jointly Gaussian because they are linear combinations of a Gaussian vector. If $i < j$, then

$$\begin{aligned}&\operatorname{Cov}(B_{t_i}^0 - B_{t_{i-1}}^0,\ B_{t_j}^0 - B_{t_{j-1}}^0)\\ &\quad = (t_i \wedge t_j) - (t_i \wedge t_{j-1}) - (t_{i-1} \wedge t_j) + (t_{i-1} \wedge t_{j-1}) = 0.\end{aligned}$$

Uncorrelated components of a jointly Gaussian vector are independent. Also $\operatorname{Var}(B_t^0 - B_s^0) = t - s$, which proves the asserted increment laws, while $\operatorname{Var}(B_0^0) = 0$ gives $B_0^0 = 0$ almost surely. □

At this stage Ω_0 contains every real-valued path, including wildly discontinuous ones. Continuity has not yet been proved.

10.2 Kolmogorov–Chentsov and the continuous modification

Theorem 10.3 (Kolmogorov–Chentsov continuity criterion)**.** *Let $(X_t)_{t\in[0,T]}$ take values in $\mathbb{R}^d$. Suppose that for some $p > 0$, $\beta > 0$, and $C < \infty$,*

$$\mathbb{E}|X_t - X_s|^p \le C|t-s|^{1+\beta} \qquad (s,t \in [0,T]). \tag{10.3}$$

Then X has a modification whose paths are Hölder continuous of every order $\alpha < \beta/p$.

Proof. It is enough to work on $[0,1]$; deterministic rescaling handles $[0,T]$. Let $D_m = \{k2^{-m} : 0 \le k \le 2^m\}$. For $0 < \alpha < \beta/p$, Markov's inequality and (10.3) give

$$\begin{aligned}\mathbb{P}\Big(\max_{0\le k<2^m} |X_{(k+1)2^{-m}} - X_{k2^{-m}}| > 2^{-m\alpha}\Big) \\ \le 2^m C\, 2^{-m(1+\beta)} 2^{mp\alpha} = C2^{-m(\beta-p\alpha)}.\end{aligned} \tag{10.4}$$

The series over m is finite. Borel–Cantelli therefore implies that, outside one null set, all sufficiently fine adjacent dyadic increments are at most $2^{-m\alpha}$. A dyadic chaining argument then yields, for every $\alpha' < \alpha$, a finite random constant $C_{\alpha'}(\omega)$ such that

$$|X_t - X_s| \le C_{\alpha'}(\omega)|t-s|^{\alpha'} \quad \text{for all dyadic } s,t. \tag{10.5}$$

Thus the restriction to the countable dense set $D = \bigcup_m D_m$ has a unique continuous extension $\widetilde{X}$. For a fixed t, choose dyadic $t_m \to t$. The moment estimate gives $X_{t_m} \to X_t$ in probability, while the construction gives $X_{t_m} \to \widetilde{X}_t$ almost surely. Hence $X_t = \widetilde{X}_t$ almost surely. This is precisely the modification property. Letting $\alpha \uparrow \beta/p$ completes the proof. □

For a centered Gaussian $G \sim N(0,v)$,

$$\mathbb{E}|G|^{2m} = (2m-1)!!\, v^m. \tag{10.6}$$

Consequently the coordinate process satisfies

$$\mathbb{E}|B^0_t - B^0_s|^{2m} = (2m-1)!!\,|t-s|^m. \tag{10.7}$$

Here $p = 2m$ and $\beta = m-1$, so every $\alpha < (m-1)/(2m)$ is available. Sending $m \to \infty$ gives one continuous modification B whose paths are locally Hölder continuous of every order $\alpha < 1/2$. The modification cannot be Hölder of order $1/2$ in general; the continuity theorem is deliberately strict at the endpoint.

Definition 10.4 (Wiener measure)**.** For $T < \infty$, the law of the continuous modification as a random element of $C_0([0,T]) = \{\omega \in C([0,T]) : \omega(0) = 0\}$, equipped with the uniform norm and its Borel σ-field, is Wiener measure W_T.

The Borel σ-field of $C_0([0,T])$ is generated by the evaluations at rational times. Therefore the Gaussian finite-dimensional distributions determine W_T uniquely. After completing the filtration and making it right-continuous, the coordinate process is standard Brownian motion on the canonical Wiener space.

Construction principle

Kolmogorov extension controls joint laws on cylinder events. The Kolmogorov–Chentsov estimate controls the geometry of almost every path. No continuity conclusion follows from the extension theorem alone.

11 The L^2 Itô integral and Itô isometry

Fix $T < \infty$ and a Brownian filtration satisfying the usual conditions. Let $\mathcal{P}$ denote the predictable σ-field on $\Omega \times [0,T]$, and define

$$\mathsf{H}_T^2 = L^2(\Omega \times [0,T], \mathcal{P}, \mathbb{P} \otimes \mathrm{d}t). \tag{11.1}$$

11.1 Elementary predictable integrands

An elementary predictable process has the form

$$H_t = \sum_{j=0}^{m-1} H_j \mathbf{1}_{(t_j, t_{j+1}]}(t), \qquad 0 = t_0 < \cdots < t_m = T, \tag{11.2}$$

where each H_j is bounded and $\mathcal{F}_{t_j}$-measurable. Define

$$I_T(H) = \int_0^T H_t \, \mathrm{d}B_t := \sum_{j=0}^{m-1} H_j (B_{t_{j+1}} - B_{t_j}). \tag{11.3}$$

Theorem 11.1 (Itô isometry for elementary processes)**.** *For elementary predictable H, K,*

$$\mathbb{E}[I_T(H)] = 0, \tag{11.4}$$

$$\mathbb{E}[I_T(H) I_T(K)] = \mathbb{E} \int_0^T H_t K_t \, \mathrm{d}t. \tag{11.5}$$

In particular,

$$\|I_T(H)\|_{L^2(\Omega)}^2 = \|H\|_{\mathsf{H}_T^2}^2. \tag{11.6}$$

Proof. Refine the two deterministic partitions to a common partition and write $\Delta_j B = B_{t_{j+1}} - B_{t_j}$. Since H_j is known at time t_j and $\mathbb{E}[\Delta_j B \mid \mathcal{F}_{t_j}] = 0$, every summand has mean zero. If $i < j$, then $H_i \Delta_i B K_j$ is $\mathcal{F}_{t_j}$-measurable, hence

$$\mathbb{E}[H_i \Delta_i B K_j \Delta_j B] = \mathbb{E}\big[H_i \Delta_i B K_j \mathbb{E}(\Delta_j B \mid \mathcal{F}_{t_j})\big] = 0.$$

Thus all off-diagonal terms vanish. On the diagonal, Brownian independence gives

$$\begin{aligned} \mathbb{E}[H_j K_j (\Delta_j B)^2] &= \mathbb{E}\big[H_j K_j \mathbb{E}((\Delta_j B)^2 \mid \mathcal{F}_{t_j})\big] \\ &= (t_{j+1} - t_j) \mathbb{E}[H_j K_j]. \end{aligned}$$

Summing proves (11.5); taking $K = H$ gives the isometry. □

11.2 Completion in L^2

Elementary predictable processes are dense in H_T^2. Indeed, indicators of predictable rectangles $A \times (s,t]$, with $A \in \mathcal{F}_s$, generate $\mathcal{P}$; the usual simple-function approximation and truncation argument then gives density. If $H \in \mathsf{H}_T^2$, choose elementary $H^{(n)}$ with $\|H^{(n)} - H\|_{\mathsf{H}_T^2} \to 0$. By (11.6), $I_T(H^{(n)})$ is Cauchy in $L^2(\Omega)$. Define

$$\int_0^T H_t \, \mathrm{d}B_t := L^2(\Omega)\text{-}\lim_{n \to \infty} I_T(H^{(n)}). \tag{11.7}$$

The limit is independent of the approximation, and continuity passes (11.5) and (11.6) to all of H_T^2.

Theorem 11.2 (Martingale and maximal properties)**.** *For $H \in \mathsf{H}_T^2$, the process*

$$M_t = \int_0^t H_s \, \mathrm{d}B_s$$

has a continuous square-integrable martingale modification, with predictable quadratic variation

$$\langle M \rangle_t = \int_0^t H_s^2 \, \mathrm{d}s, \tag{11.8}$$

and

$$\mathbb{E} \sup_{0 \le t \le T} |M_t|^2 \le 4 \mathbb{E} \int_0^T H_s^2 \, \mathrm{d}s. \tag{11.9}$$

Proof. For elementary integrands the martingale property follows by conditioning each future Brownian increment on the past. Approximate H in H^2_T. Doob's L^2 maximal inequality and the isometry give

$$\mathbb{E}\sup_{t\le T}|I_t(H^{(n)})-I_t(H^{(m)})|^2\le 4\|H^{(n)}-H^{(m)}\|^2_{\mathsf{H}^2_T}.$$

Hence the continuous elementary integrals converge uniformly in probability, and along a subsequence uniformly almost surely, to a continuous martingale version. Applying the same approximation to $M_t^2-\int_0^t H_s^2\,\mathrm{d}s$ gives (11.8). The displayed maximal estimate follows by taking limits. □

Remark 11.3 (Isometry versus isomorphism)**.** The map $H\mapsto\int_0^T H\,\mathrm{d}B$ is an isometric linear embedding of H^2_T into the centered part of $L^2(\mathcal{F}_T)$. Surjectivity is a separate martingale-representation theorem: for the augmented natural Brownian filtration, every $Y\in L^2(\mathcal{F}_T)$ has a unique representation $Y=\mathbb{E}Y+\int_0^T H_s\,\mathrm{d}B_s$.

11.3 Itô's formula as the stochastic chain rule

Theorem 11.4 (Itô formula)**.** *Let*

$$X_t=X_0+\int_0^t b_s\,\mathrm{d}s+\int_0^t\sigma_s\,\mathrm{d}B_s,$$

where the integrals are well-defined, and let $f\in C^{1,2}([0,T]\times\mathbb{R})$. After localization if necessary,

$$\begin{aligned}f(t,X_t)=f(0,X_0)&+\int_0^t\left(\partial_s f+b_s\partial_x f+\frac12\sigma_s^2\partial_{xx}f\right)(s,X_s)\,\mathrm{d}s\\&+\int_0^t\sigma_s\partial_x f(s,X_s)\,\mathrm{d}B_s.\end{aligned}\tag{11.10}$$

Proof. First assume all coefficients and the derivatives of f are bounded. On a partition $0=t_0<\cdots<t_n=t$, apply the second-order Taylor formula to $f(t_{j+1},X_{t_{j+1}})-f(t_j,X_{t_j})$. Summation produces the time increments, the first-order increments of X, the quadratic term, and a remainder. As the mesh tends to zero, the finite-variation squares and the cross products with Brownian increments vanish in probability, while

$$\sum_j(\Delta B_j)^2\longrightarrow t\quad\text{and}\sum_j\partial_x f(t_j,X_{t_j})\sigma_{t_j}\Delta B_j\longrightarrow\int_0^t\partial_x f\,\sigma\,\mathrm{d}B$$

in probability, the latter by Itô isometry. Uniform continuity of the second derivatives on compact sets makes the Taylor remainder vanish. This yields (11.10) for bounded data. Stopping when $|X|+\int(|b|+\sigma^2)$ exceeds an integer and then letting the integer tend to infinity proves the localized statement. □

For $f(x)=e^{\theta x}$ and $X_t=\mu t+\sigma B_t$, (11.10) immediately recovers the compensator $\mu\theta+\frac12\sigma^2\theta^2$ used in Section 4.

12 A complete path-space proof of Donsker's theorem

Let $\xi_1, \xi_2, \dots$ be independent Rademacher variables and, on $[0,T]$, let

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} \left(S_{\lfloor nt \rfloor} + (nt - \lfloor nt \rfloor) \xi_{\lfloor nt \rfloor + 1} \right). \tag{12.1}$$

Write P_n for its law on the Polish space $(C_0([0,T]), \|\cdot\|_\infty)$.

12.1 Finite-dimensional convergence

Fix $0 = t_0 < t_1 < \cdots < t_m \le T$. The interpolation error at any fixed time is bounded by $n^{-1/2}$. For the grid-point increments, independence gives

$$\begin{aligned} &\mathbb{E} \exp\left(i \sum_{j=1}^{m} u_j \left[W^{(n)}(t_j) - W^{(n)}(t_{j-1}) \right] \right) \\ &= \prod_{j=1}^{m} \left(\cos \frac{u_j}{\sqrt{n}} \right)^{\lfloor nt_j \rfloor - \lfloor nt_{j-1} \rfloor} \longrightarrow \prod_{j=1}^{m} e^{-u_j^2 (t_j - t_{j-1})/2}. \end{aligned} \tag{12.2}$$

The limit is the joint characteristic function of independent Gaussian increments. Lévy's continuity theorem and the invertible linear map from increments to levels therefore yield

$$(W^{(n)}(t_1), \dots, W^{(n)}(t_m)) \Rightarrow (B_{t_1}, \dots, B_{t_m}). \tag{12.3}$$

12.2 The uniform fourth-moment estimate

For $0 \le s < t \le T$, let ℓ_j be the length of the intersection of $(s,t]$ with the j-th grid cell $(j/n, (j+1)/n]$. Linear interpolation gives the exact identity

$$W^{(n)}(t) - W^{(n)}(s) = \sum_j a_j \xi_{j+1}, \qquad a_j = \sqrt{n}\, \ell_j. \tag{12.4}$$

Since $0 \le \ell_j \le 1/n$ and $\sum_j \ell_j = t - s$,

$$\sum_j a_j^2 = n \sum_j \ell_j^2 \le t - s. \tag{12.5}$$

For independent Rademacher variables,

$$\mathbb{E} \left(\sum_j a_j \xi_j \right)^4 = 3 \left(\sum_j a_j^2 \right)^2 - 2 \sum_j a_j^4 \le 3 \left(\sum_j a_j^2 \right)^2. \tag{12.6}$$

Combining the last three displays yields the estimate, uniform in n,

$$\boxed{\mathbb{E} |W^{(n)}(t) - W^{(n)}(s)|^4 \le 3 |t - s|^2.} \tag{12.7}$$

12.3 Dyadic modulus control and Arzelà–Ascoli

Fix $0 < \alpha < 1/4$ and use the dyadic grid $D_r = \{kT2^{-r} : 0 \le k \le 2^r\}$. Define

$$A_\alpha(x) = \sup_{r \ge 0} 2^{r\alpha} \max_{0 \le k < 2^r} |x((k+1)T2^{-r}) - x(kT2^{-r})|. \tag{12.8}$$

By the union bound, Markov's inequality, and (12.7),

$$\begin{aligned} \sup_n \mathbb{P}(A_\alpha(W^{(n)}) > R) &\le \sum_{r=0}^{\infty} 2^r \frac{3 (T 2^{-r})^2}{R^4 2^{-4r\alpha}} \\ &= \frac{3T^2}{R^4} \sum_{r=0}^{\infty} 2^{-r(1-4\alpha)} \xrightarrow[R \to \infty]{} 0. \end{aligned} \tag{12.9}$$

Set

$$K_R = \{x \in C_0([0,T]) : A_\alpha(x) \le R\}.$$

Dyadic chaining shows that every $x \in K_R$ satisfies

$$|x(t) - x(s)| \le C_{\alpha,T} R |t - s|^\alpha \qquad (s,t \in [0,T]), \tag{12.10}$$

with a deterministic constant. In particular, K_R is uniformly bounded and equicontinuous. It is also closed, since every dyadic increment inequality is preserved by uniform limits. Arzelà–Ascoli therefore makes K_R compact. Equation (12.9) proves

$$\forall \varepsilon > 0 \; \exists R < \infty : \inf_n \mathsf{P}_n(K_R) \geq 1 - \varepsilon; \tag{12.11}$$

that is, (P_n) is tight in $C_0([0,T])$.

12.4 Prokhorov plus uniqueness of the subsequential limit

Theorem 12.1 (Prokhorov theorem on a Polish space)**.** *A family of probability measures on a Polish space is tight if and only if it is relatively compact for weak convergence.*

By Prokhorov, every subsequence of (P_n) has a weakly convergent subsequence. If $\mathsf{P}_{n_k} \Rightarrow \mathsf{P}$, then the evaluation map

$$x \longmapsto (x(t_1), \ldots, x(t_m))$$

is continuous on $C_0([0,T])$. The continuous mapping theorem and (12.3) imply that P has Brownian finite-dimensional distributions. Evaluations at rational times generate the Borel σ-field, so $\mathsf{P} = \mathsf{W}_T$. Every convergent subsequence has the same limit; consequently the whole sequence converges:

$$W^{(n)} \Rightarrow B \qquad \text{in } C_0([0,T]). \tag{12.12}$$

This closes the tightness gap left by one-time characteristic functions.

For the weakly biased walk of (2.16), subtract the deterministic mean μt. Its centered increments form a bounded triangular array with variance $\sigma^2\delta + O(\delta^2)$ and satisfy the Lindeberg condition automatically. The joint-CF computation gives the drifted Gaussian finite-dimensional laws, while the centered fourth-moment bound has the same form as (12.7), up to a constant independent of δ. The preceding tightness argument therefore gives

$$X^\delta \Rightarrow (t \mapsto \mu t + \sigma B_t) \quad \text{in } C([0,T]). \tag{12.13}$$

Where each theorem enters

Lévy continuity identifies every finite-dimensional Gaussian limit. Arzelà–Ascoli identifies compact subsets of path space. The fourth-moment estimate puts almost all path laws inside such compact sets. Prokhorov then extracts path-space limits, and uniqueness of Brownian finite-dimensional laws identifies them.

13 Girsanov change of measure and drift calibration

The exponential martingales above have two distinct uses. One may choose a root of the cumulant to construct a harmonic function for optional stopping, or normalize an exponential to obtain a Radon–Nikodym density and change the law of the paths. These choices need not coincide.

13.1 The constant-drift theorem with a complete calculation

Fix $T < \infty$ and $\lambda \in \mathbb{R}$. Set

$$Z_t = \exp\left(-\lambda B_t - \frac{1}{2}\lambda^2 t\right), \qquad 0 \le t \le T. \tag{13.1}$$

Independent Gaussian increments show that $\mathbb{E}[Z_t \mid \mathcal{F}_s] = Z_s$; hence Z is a positive mean-one martingale. Define a probability measure $\mathbb{Q}$ on $\mathcal{F}_T$ by

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T. \tag{13.2}$$

Theorem 13.1 (Constant-drift Girsanov theorem)**.** *Under $\mathbb{Q}$, the process*

$$W_t = B_t + \lambda t \tag{13.3}$$

is standard Brownian motion on $[0, T]$.

Proof. Continuity is unchanged by an equivalent measure. For $0 \le s < t \le T$, Bayes' formula and $Z_t/Z_s = exp(-\lambda\Delta B - lambda^2(t-s)/2)$ give, for every $u \in \mathbb{R}$,

$$\begin{aligned}
&\mathbb{E}_{\mathbb{Q}}\left[e^{iu(W_t - W_s)} \mid \mathcal{F}_s\right] \\
&\quad = \frac{1}{Z_s}\mathbb{E}_{\mathbb{P}}\left[Z_t e^{iu(\Delta B + \lambda(t-s))} \mid \mathcal{F}_s\right] \\
&\quad = e^{iu\lambda h - \lambda^2 h/2}\mathbb{E}_{\mathbb{P}}[e^{(iu-\lambda)\Delta B}] = e^{-u^2 h/2}, \qquad h = t - s.
\end{aligned}$$

The last quantity is deterministic, so $W_t - W_s$ is independent of $\mathcal{F}_s$ under $\mathbb{Q}$, and it has law $N(0, t-s)$. Thus W is Brownian motion. □

For $X_t = x_0 + \mu t + \sigma B_t$, take $\lambda = \mu/\sigma$. Then under $\mathbb{Q}$,

$$X_t = x_0 + \sigma W_t, \qquad \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \exp\left(-\frac{\mu}{\sigma}B_T - \frac{1}{2}\frac{\mu^2}{\sigma^2}T\right). \tag{13.4}$$

Conversely, on canonical path space, if $\mathbb{P}^0$ is the zero-drift law and $\mathbb{P}^\mu$ is the law with drift μ, then

$$\left.\frac{\mathrm{d}\mathbb{P}^\mu}{\mathrm{d}\mathbb{P}^0}\right|_{\mathcal{F}_T} = \exp\left(\frac{\mu}{\sigma}B_T - \frac{1}{2}\frac{\mu^2}{\sigma^2}T\right). \tag{13.5}$$

13.2 General predictable drift

Theorem 13.2 (Girsanov theorem under Novikov's condition)**.** *Let θ be predictable and satisfy*

$$\mathbb{E}\exp\left(\frac{1}{2}\int_0^T \theta_s^2\,\mathrm{d}s\right) < \infty. \tag{13.6}$$

Then the stochastic exponential

$$Z_t = \exp\left(-\int_0^t \theta_s\,\mathrm{d}B_s - \frac{1}{2}\int_0^t \theta_s^2\,\mathrm{d}s\right) \tag{13.7}$$

is a uniformly integrable martingale with $\mathbb{E}Z_T = 1$. Under $\mathrm{d}\mathbb{Q} = Z_T\,\mathrm{d}\mathbb{P}$,

$$W_t = B_t + \int_0^t \theta_s\,\mathrm{d}s \tag{13.8}$$

is Brownian motion on $[0, T]$.

Proof. Itô's formula gives $\mathrm{d}Z_t = -Z_t\theta_t\,\mathrm{d}B_t$, so Z is a positive local martingale. Novikov's exponential-integrability condition is the standard uniform-integrability criterion obtained by stopping when $\int\theta^2$ or Z becomes large and applying Hölder's inequality to the stopped stochastic exponentials; it upgrades Z to a true mean-one martingale.

It remains to identify the process after the measure change. Since $[Z, W]_t = -\int_0^t Z_s\theta_s \, \mathrm{d}s$, integration by parts gives

$$\begin{aligned} \mathrm{d}(Z_tW_t) &= W_t \, \mathrm{d}Z_t + Z_t \, \mathrm{d}W_t + \mathrm{d}[Z, W]_t \\ &= Z_t(1 - W_t\theta_t) \, \mathrm{d}B_t. \end{aligned}$$

Thus ZW is a $\mathbb{P}$-local martingale. Bayes' formula, first for stopped bounded processes and then by localization, implies that W is a $\mathbb{Q}$-local martingale. Its quadratic variation is pathwise $[W]_t = [B]_t = t$. Lévy's martingale characterization therefore makes W a $\mathbb{Q}$-Brownian motion. □

Remark 13.3 (The optional-stopping root flips rather than removes drift)**.** The nonzero root used earlier is $\theta_* = -2\mu/\sigma^2$. Its uncorrected mean-one exponential can be written

$$e^{\theta_*(X_t - x_0)} = \exp\left(-\frac{2\mu}{\sigma}B_t - 2\frac{\mu^2}{\sigma^2}t\right).$$

As a density it corresponds to $\lambda = 2\mu/\sigma$, so it changes the drift of X from $+\mu$ to $-\mu$. The drift-removing density is (13.4). Harmonic calibration and drift removal are therefore related exponential operations with different parameters.

14 Generator, Feynman–Kac, and weak PDE solutions

Consider the time-homogeneous diffusion

$$\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}B_t, \qquad X_0 = x, \tag{14.1}$$

under conditions ensuring a nonexplosive solution. Write $P_t f(x) = \mathbb{E}_x[f(X_t)]$ for its Markov semigroup.

14.1 Infinitesimal generator and Dynkin's formula

Definition 14.1 (Infinitesimal generator)**.** For functions for which the limit exists,

$$\mathcal{L}f(x) = \lim_{t \downarrow 0} \frac{P_t f(x) - f(x)}{t}.$$

For $f \in C_c^2(\mathbb{R})$, Itô's formula identifies

$$\mathcal{L}f(x) = b(x) f'(x) + \frac{1}{2}\sigma^2(x) f''(x). \tag{14.2}$$

Indeed, stopping inside a compact set, taking expectations in Itô's formula, and dividing by t gives

$$\frac{P_t f(x) - f(x)}{t} = \frac{1}{t}\mathbb{E}_x \int_0^t \mathcal{L}f(X_s)\,\mathrm{d}s \longrightarrow \mathcal{L}f(x)$$

by path continuity and dominated convergence.

Theorem 14.2 (Dynkin formula)**.** *If τ is a bounded stopping time and $f \in C_c^2(\mathbb{R})$, then*

$$\mathbb{E}_x[f(X_\tau)] - f(x) = \mathbb{E}_x \int_0^\tau \mathcal{L}f(X_s)\,\mathrm{d}s. \tag{14.3}$$

Proof. Apply Itô's formula to $f(X_{t\wedge\tau})$. The stopped stochastic integral has expectation zero by Itô isometry and localization. Let $t \uparrow T$, where $\tau \le T$, and use dominated convergence. □

For constant $b = \mu$ and $\sigma > 0$,

$$\mathcal{L} = \mu\partial_x + \frac{1}{2}\sigma^2\partial_{xx}, \qquad \mathcal{L}^* = -\mu\partial_x + \frac{1}{2}\sigma^2\partial_{xx}, \tag{14.4}$$

where $\mathcal{L}^*$ is the formal adjoint acting on densities.

14.2 Backward equation and Feynman–Kac

Theorem 14.3 (Feynman–Kac formula)**.** *Let b, σ be bounded Lipschitz functions with $\sigma^2 \ge c > 0$, and let V, f, g be bounded and continuous. If $u \in C^{1,2}([0,T)\times\mathbb{R}) \cap C([0,T]\times\mathbb{R})$ is a bounded classical solution of*

$$\begin{aligned} \partial_t u + \mathcal{L}u - Vu + f &= 0, && 0 \le t < T, \\ u(T,x) &= g(x), \end{aligned} \tag{14.5}$$

then

$$u(t,x) = \mathbb{E}_{t,x}\left[e^{-\int_t^T V(r,X_r)\,\mathrm{d}r} g(X_T) + \int_t^T e^{-\int_t^s V(r,X_r)\,\mathrm{d}r} f(s,X_s)\,\mathrm{d}s \right]. \tag{14.6}$$

Conversely, fix $0 < \alpha < 1$. If in addition $b, \sigma \in C_b^\alpha(\mathbb{R})$, $V, f \in C_b^{\alpha/2,\alpha}([0,T]\times\mathbb{R})$, and $g \in C_b^{2+\alpha}(\mathbb{R})$, then the right-hand side belongs locally to the parabolic class $C^{1+\alpha/2,2+\alpha}([0,T)\times\mathbb{R})$, extends continuously to the terminal data, and is the unique bounded classical solution.

Proof. Fix (t,x), let $X_t = x$, and put $A_s = \int_t^s V(r,X_r)\,\mathrm{d}r$. Apply Itô's product rule to

$$Y_s = e^{-A_s} u(s,X_s) + \int_t^s e^{-A_r} f(r,X_r)\,\mathrm{d}r.$$

The drift of Y is

$$e^{-A_s}(\partial_s u + \mathcal{L}u - Vu + f)(s,X_s)\,\mathrm{d}s = 0,$$

and its remaining term is $e^{-A_s}\sigma(X_s)\partial_x u(s,X_s)\,\mathrm{d}B_s$. After localization this is a true martingale because the data are bounded. Hence $u(t,x) = Y_t = \mathbb{E}_{t,x}Y_T$, which is (14.6). For the converse, the Markov property first gives the dynamic programming identity. Parabolic interior regularity then gives $C^{1,2}$ smoothness, and applying the generator to the short-time identity recovers (14.5). The forward implication applied to the difference of two solutions proves uniqueness. □

When $V = f = 0$, this reduces to the backward Kolmogorov equation

$$\partial_t u + \mathcal{L}u = 0, \qquad u(T,\cdot) = g, \qquad u(t,x) = P_{T-t}g(x). \tag{14.7}$$

14.3 Forward equation as a weak identity

Let $\mu_t = \mathrm{Law}(X_t)$. Applying Itô's formula to a test function $\varphi \in C_c^\infty(\mathbb{R})$, taking expectations, and using Fubini gives

$$\int \varphi\,\mathrm{d}\mu_t - \int \varphi\,\mathrm{d}\mu_0 = \int_0^t \int \mathcal{L}\varphi(x)\mu_s(\,\mathrm{d}x)\,\mathrm{d}s. \tag{14.8}$$

This is the weak Fokker–Planck equation

$$\partial_t \mu_t = \mathcal{L}^*\mu_t. \tag{14.9}$$

If μ_t has density $\rho(t,x)$, then for every $\eta \in C_c^\infty([0,T)\times\mathbb{R})$,

$$\int_0^T \int_{\mathbb{R}} \rho(t,x)(\partial_t\eta + \mathcal{L}\eta)(t,x)\,\mathrm{d}x\,\mathrm{d}t + \int_{\mathbb{R}} \eta(0,x)\mu_0(\,\mathrm{d}x) = 0. \tag{14.10}$$

Equation (14.10) is meaningful even when ρ has no classical derivatives. It is the appropriate distributional notion of a forward solution.

For Brownian motion, $\mathcal{L} = \frac{1}{2}\partial_{xx}$, and the heat semigroup is

$$P_t f(x) = p_t * f(x), \qquad p_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}. \tag{14.11}$$

For bounded measurable f, differentiation of the Gaussian kernel gives

$$P_t f \in C^\infty(\mathbb{R}) \quad (t>0), \qquad \|\partial_x^k P_t f\|_\infty \le C_k t^{-k/2}\|f\|_\infty. \tag{14.12}$$

Thus the semigroup instantly regularizes rough initial data. For a finite initial measure μ_0, $\rho(t,\cdot) = p_t * \mu_0$ is smooth for every $t>0$, satisfies (14.10), and converges weakly to μ_0 as $t \downarrow 0$.

Proposition 14.4 (Uniqueness of the weak heat flow)**.** *Among narrowly continuous finite-measure curves with uniformly bounded total mass, $\rho_t = p_t * \mu_0$ is the unique solution of (14.10) for $\mathcal{L} = \frac{1}{2}\partial_{xx}$.*

Proof. Fix $t>0$ and $\varphi \in C_c^\infty(\mathbb{R})$. For $0 \le s < t$, set $v(s,x) = P_{t-s}\varphi(x)$. Then $\partial_s v + \frac{1}{2}\partial_{xx}v = 0$. Insert a time-cutoff version of v into the weak identity and let the cutoff approach the endpoints. One obtains

$$\int \varphi\,\mathrm{d}\mu_t = \int P_t\varphi\,\mathrm{d}\mu_0.$$

The right-hand side is exactly integration against $p_t * \mu_0$. Since test functions determine finite measures, uniqueness follows. □

Probabilistic–analytic dictionary

The Itô integral supplies martingales; Itô's formula supplies the generator; the generator gives Dynkin's formula and the backward PDE; duality gives the weak forward PDE; the heat kernel upgrades weak initial data to smooth positive time solutions; and Feynman–Kac adds killing and forcing through pathwise weights.

15 Dependency map of the complete argument

The discrete spine of the argument is

$$\begin{aligned}\xi_j \text{ Rademacher} &\Longrightarrow \phi_\xi(u) = \cos u, \quad M_\xi(k) = \cosh k \\ &\Longrightarrow \mathbb{E}[S_n] = 0, \quad \operatorname{Var}(S_n) = n, \quad [S]_n = \langle S\rangle_n = n \\ &\Longrightarrow \frac{e^{\theta S_n}}{(\cosh\theta)^n} \text{ is a martingale} \\ &\xRightarrow{\Delta x = \sigma\sqrt{\Delta t}} e^{\theta\sigma B_t - \theta^2\sigma^2 t/2}.\end{aligned}$$

At a stopping boundary, the same three martingales give

$$r^{S_n}, \quad S_n, \quad S_n^2 - n \xRightarrow{\text{OST}} \mathbb{P}_0(\tau_b < \tau_{-a}) \quad \text{and} \quad \mathbb{E}_0[\tau] = ab \text{ in the fair case.}$$

Weak bias then inserts the drift:

$$\begin{aligned}p_\delta = \frac{1}{2}\left(1 + \frac{\mu}{\sigma}\sqrt{\delta}\right) &\Longrightarrow \phi_{X_t^\delta}(u) \longrightarrow e^{i\mu u t - \sigma^2 u^2 t/2} \\ &\xRightarrow{\text{Lévy}} X_t^\delta \Rightarrow N(\mu t, \sigma^2 t) \xRightarrow{\text{tightness}} X^\delta \Rightarrow \mu\,\mathrm{id} + \sigma B.\end{aligned}$$

The measure-construction spine is

$$\begin{aligned}\Sigma_{ij} = t_i \wedge t_j &\Longrightarrow \text{consistent Gaussian finite-dimensional laws} \\ &\xRightarrow{\text{Kolmogorov extension}} \text{a coordinate process on } \mathbb{R}^{[0,\infty)} \\ &\xRightarrow{\mathbb{E}|B_t - B_s|^{2m} = C_m|t-s|^m} \text{a continuous modification} \\ &\Longrightarrow \text{Wiener measure on } C_0([0,T]).\end{aligned}$$

The functional-limit spine makes the formerly implicit tightness step explicit:

$$\begin{aligned}\mathbb{E}|W^{(n)}(t) - W^{(n)}(s)|^4 \le 3|t-s|^2 &\Longrightarrow \text{uniform dyadic modulus control} \\ &\xRightarrow{\text{Arzelà–Ascoli}} \text{compact path sets carrying mass } 1-\varepsilon \\ &\xRightarrow{\text{Prokhorov}} \text{path-space subsequential limits} \\ &\xRightarrow{\text{finite-dimensional uniqueness}} W^{(n)} \Rightarrow B \text{ in } C_0([0,T]).\end{aligned}$$

The compactness spine is

$$\begin{aligned}\text{tight laws} &\Longrightarrow \text{equicontinuous characteristic functions} \\ &\Longrightarrow \text{locally uniform CF convergence,}\end{aligned}$$

$$\phi_n(t) \to \phi(t), \quad \phi \text{ continuous at } 0 \xRightarrow{\text{Lévy}} \text{a probability weak limit with no escaped mass.}$$

Neither characteristic functions nor weak convergence encode the coupling of random variables on a common sample space.

The arithmetic side branch is

$$\rho^{-1} \text{ Pisot} \Longrightarrow \text{geometric CF resonance}, \qquad \rho^{-1} \text{ non-Pisot} \xRightarrow{\text{Erdős–Salem}} \Phi_\rho(t) \to 0.$$

Once the Brownian limit is available, the exponential-calibration chain is

$$\begin{aligned}B_t \sim N(0,t) &\Longrightarrow \mathbb{E}[e^{kB_t}] = e^{k^2 t/2} \\ &\xRightarrow{k=\theta\sigma} \mathbb{E}[e^{\theta X_t}] = e^{\theta x_0 + \psi(\theta)t} \\ &\Longrightarrow \mathcal{E}_t(\theta) = e^{\theta(X_t - x_0) - \psi(\theta)t} \text{ is a martingale} \\ &\xRightarrow{\psi(\theta_*)=0} e^{\theta_* X_t} \text{ yields two-sided exit probabilities,}\end{aligned}$$

whereas the Radon–Nikodym chain is

$$\begin{aligned}Z_T &= \exp\left(-\int_0^T \theta_s\,\mathrm{d}B_s - \frac{1}{2}\int_0^T \theta_s^2\,\mathrm{d}s\right), \\ \mathrm{d}\mathbb{Q} = Z_T\,\mathrm{d}\mathbb{P} &\xRightarrow{\text{Girsanov}} B_t + \int_0^t \theta_s\,\mathrm{d}s \text{ is Brownian under } \mathbb{Q}.\end{aligned}$$

while the imaginary substitution $k = i\omega$ gives the parallel analytic chain

$$\begin{aligned}\mathbb{E}[e^{i\omega B_t}] = e^{-t\omega^2/2} &\Longrightarrow \text{Fourier inversion}\\ &\Longrightarrow \text{completion of the square}\\ &\Longrightarrow \text{zero-residue contour shift}\\ &\Longrightarrow p_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}.\end{aligned}$$

Finally, stochastic integration closes the probabilistic–PDE circuit:

$$\begin{aligned}\text{predictable simple processes} &\xrightarrow{L^2 \text{ completion}} \int H\,\mathrm{d}B, qquad\mathbb{E}\left|\int H\,\mathrm{d}B\right|^2 = \mathbb{E}\int H^2\,\mathrm{d}t\\ &\xrightarrow{\text{Itô formula}} \mathcal{L} = b\partial_x + \tfrac{1}{2}\sigma^2\partial_{xx}\\ &\Longrightarrow \text{Dynkin and Feynman–Kac}\\ &\Longrightarrow \partial_t \mu_t = \mathcal{L}^*\mu_t \text{ in the distributional sense.}\end{aligned}$$

Final calibration principle

The substitution $k = \theta\sigma$ measures the exact coefficient with which the volatility-scaled Brownian noise enters an exponential. Balancing its Gaussian cumulant against drift produces the harmonic exponential used by optional stopping. Normalizing a possibly time-dependent exponential instead produces a Radon–Nikodym density and Girsanov's change of measure. The former solves a boundary-value problem; the latter changes the path law. Itô's formula reveals both as manifestations of the same generator and quadratic variation.